

Marine Microbial Biodiversity

Lecture 01

Microbiology 1.0.1 - The Basics

Matteo Selci, PhD

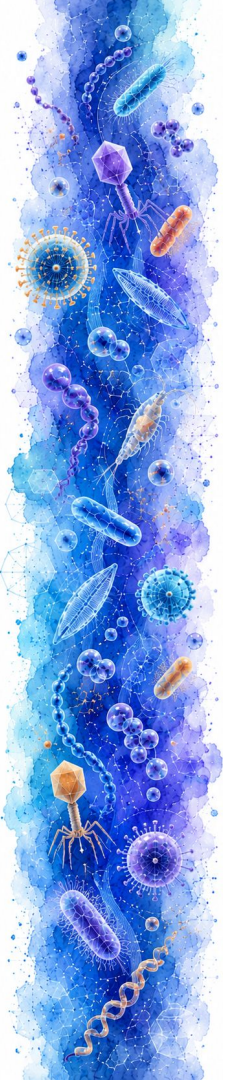
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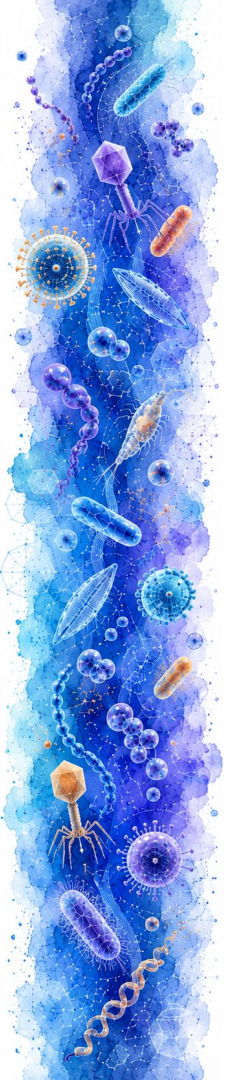
@matteo.selci

@matteo_selci





WHY STUDY MICROBIOLOGY?



Why Study Microbiology?

We can study microbes for several reasons:

- As model to understand the Origin and Evolution of Life on Earth
- As model to understanding Basic Life Principles
- As model in the search for Extraterrestrial Life
- For their contribution to our Planet Functioning
- For their Biotechnological Applications

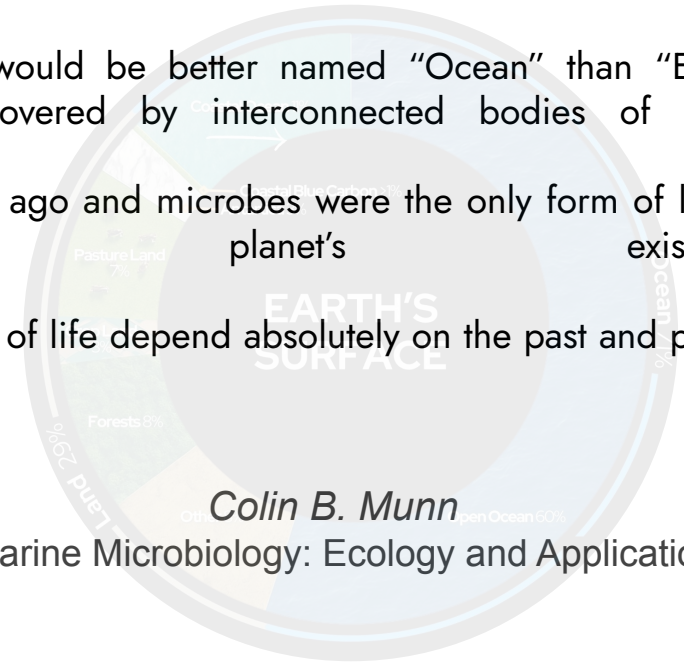
Origin and evolution of Life on Earth

“Viewed from space, it is clear why our planet would be better named “Ocean” than “Earth”. More than 70% of the planet’s surface is covered by interconnected bodies of water.

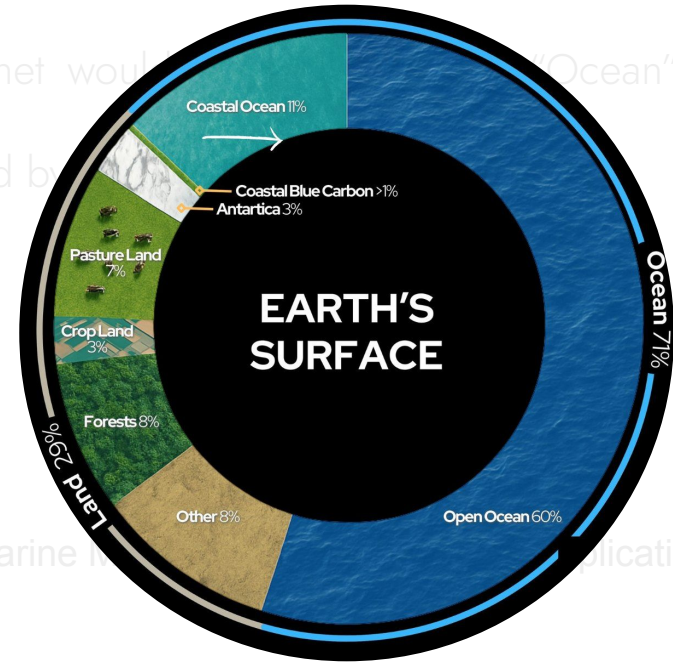
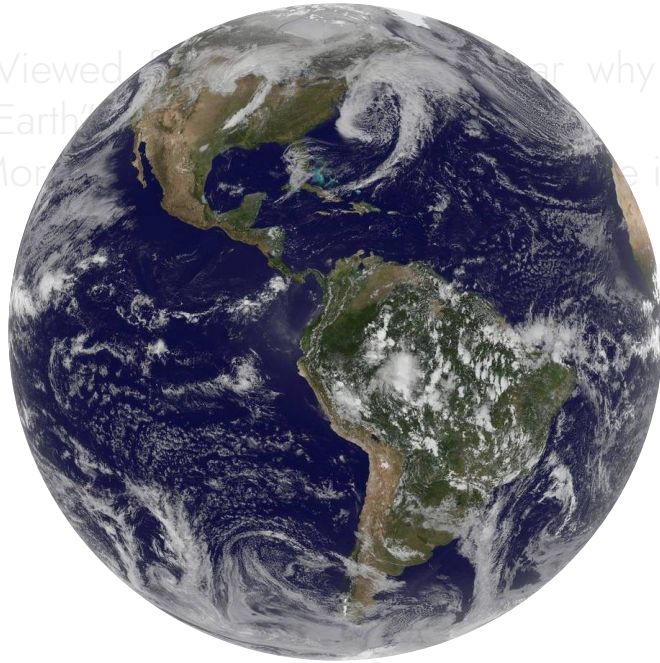
Life originated in the oceans about 3.5 billion years ago and microbes were the only form of life for two-thirds of the planet’s existence.

The development and maintenance of all other forms of life depend absolutely on the past and present activities of marine microbes.”

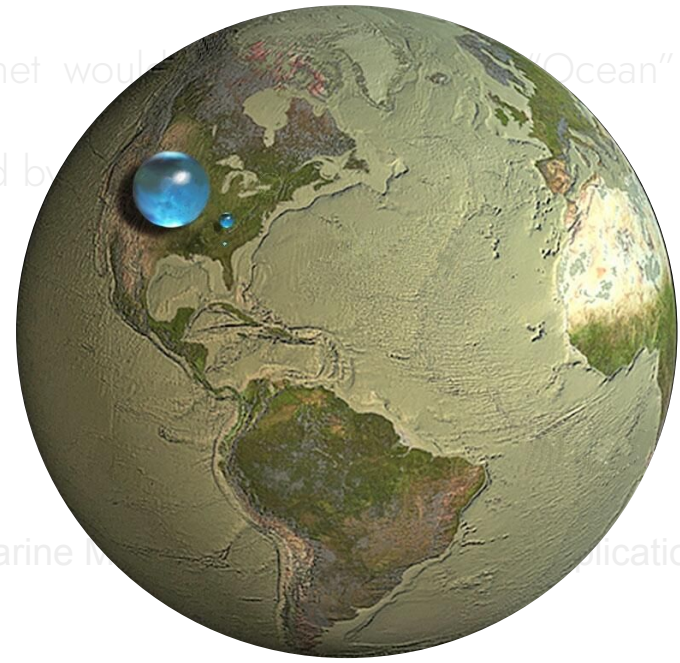
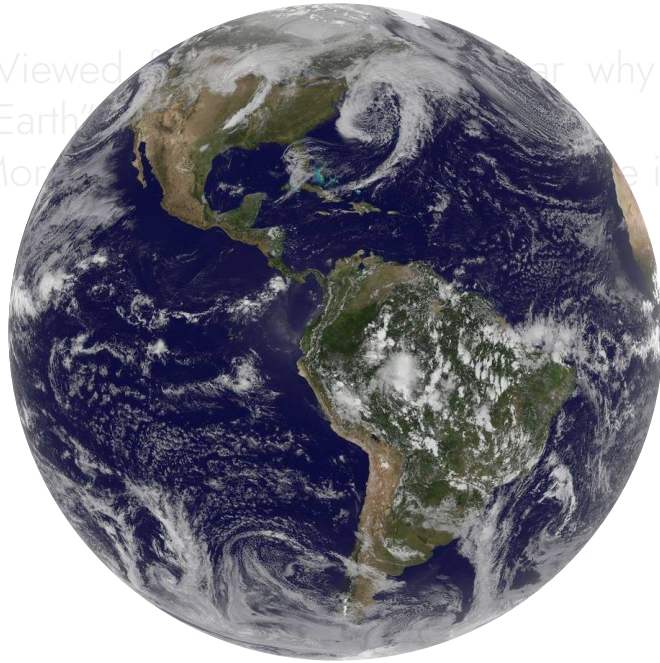
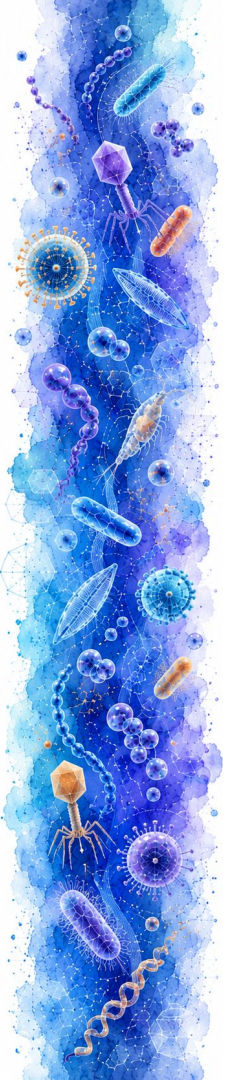
Colin B. Munn
in Marine Microbiology: Ecology and Applications



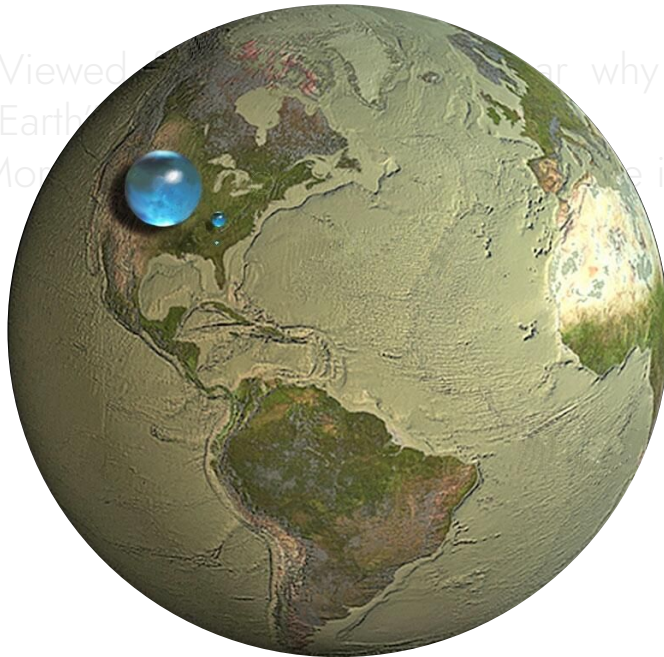
Origin and evolution of Life on Earth



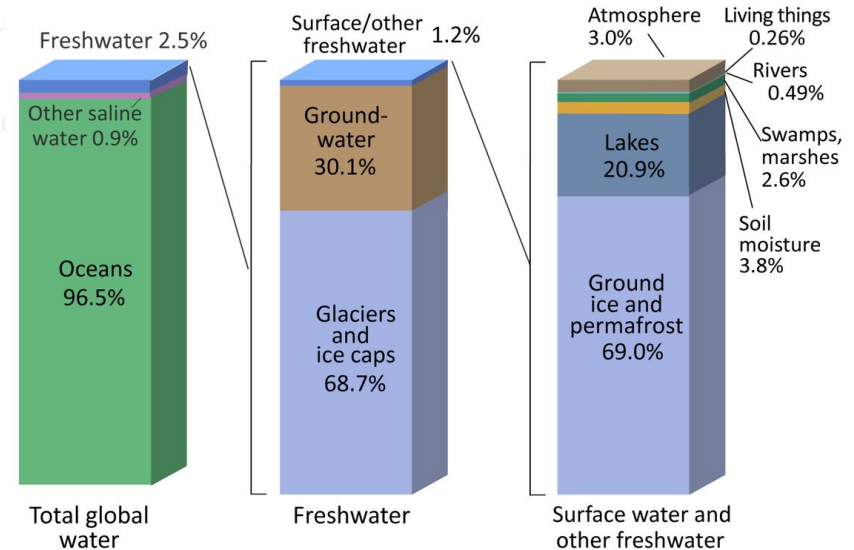
Origin and evolution of Life on Earth



Origin and evolution of Life on Earth

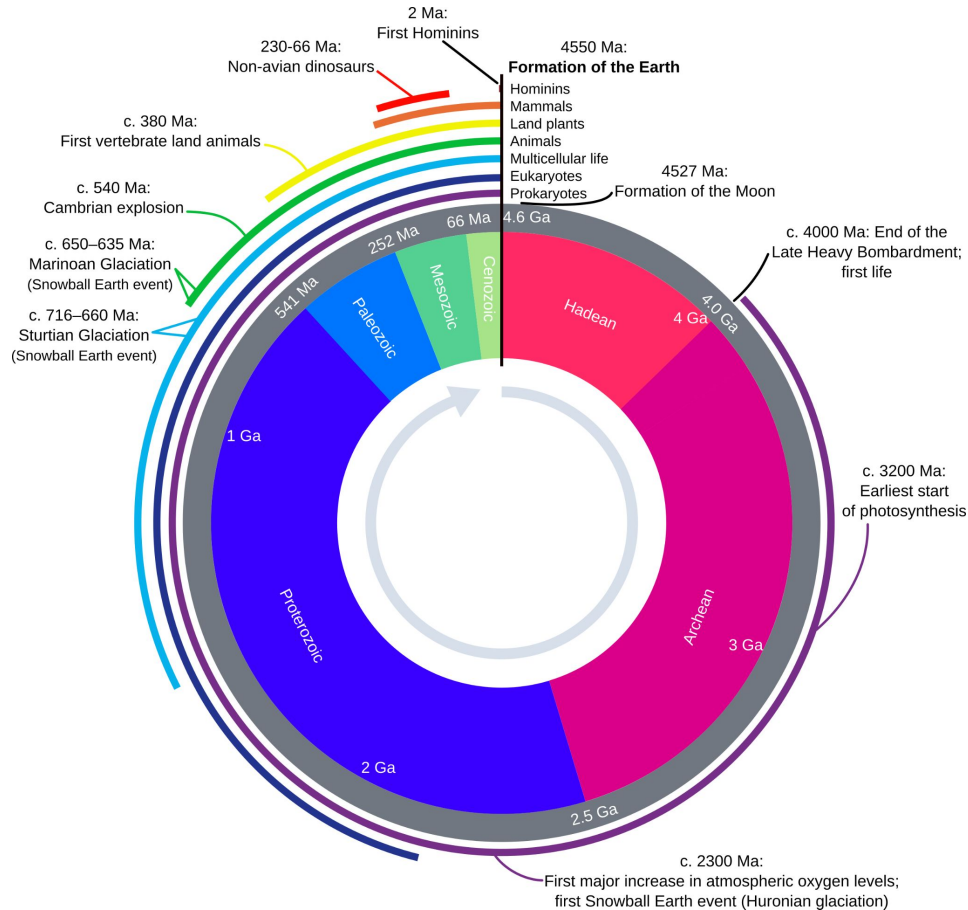


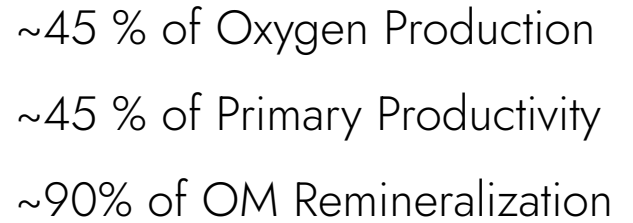
Where is Earth's Water?



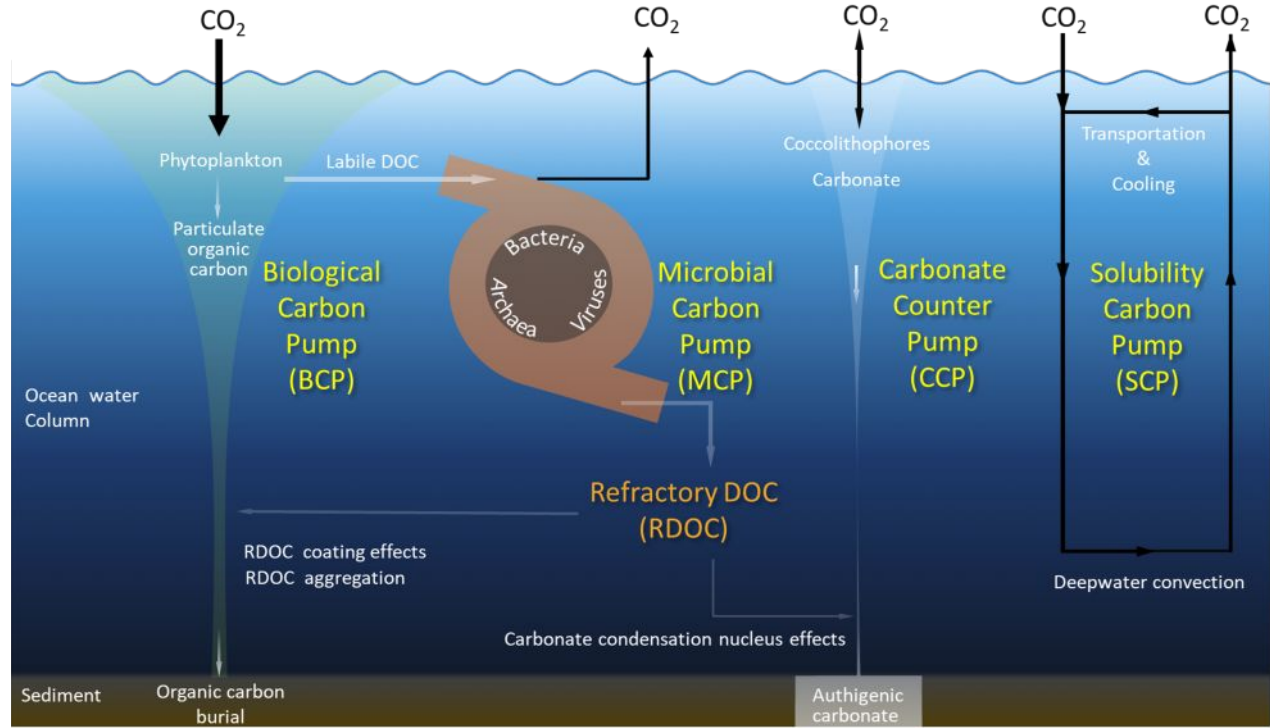
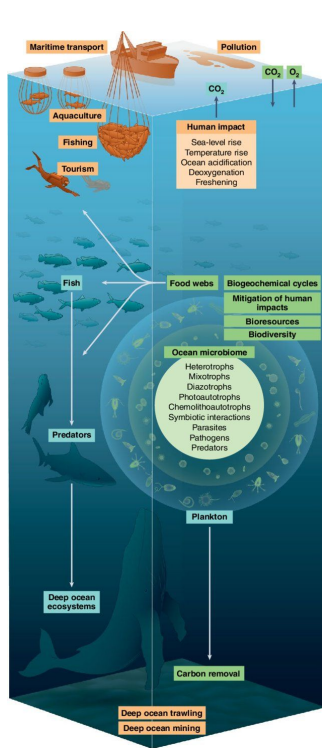
Credit: U.S. Geological Survey, Water Science School. <https://www.usgs.gov/special-topic/water-science-school>
Data source: Igor Shiklomanov's chapter "World fresh water resources" in Peter H. Gleick (editor), 1993, Water in Crisis: A Guide to the World's Fresh Water Resources. (Numbers are rounded).

Origin and evolution of Life on Earth

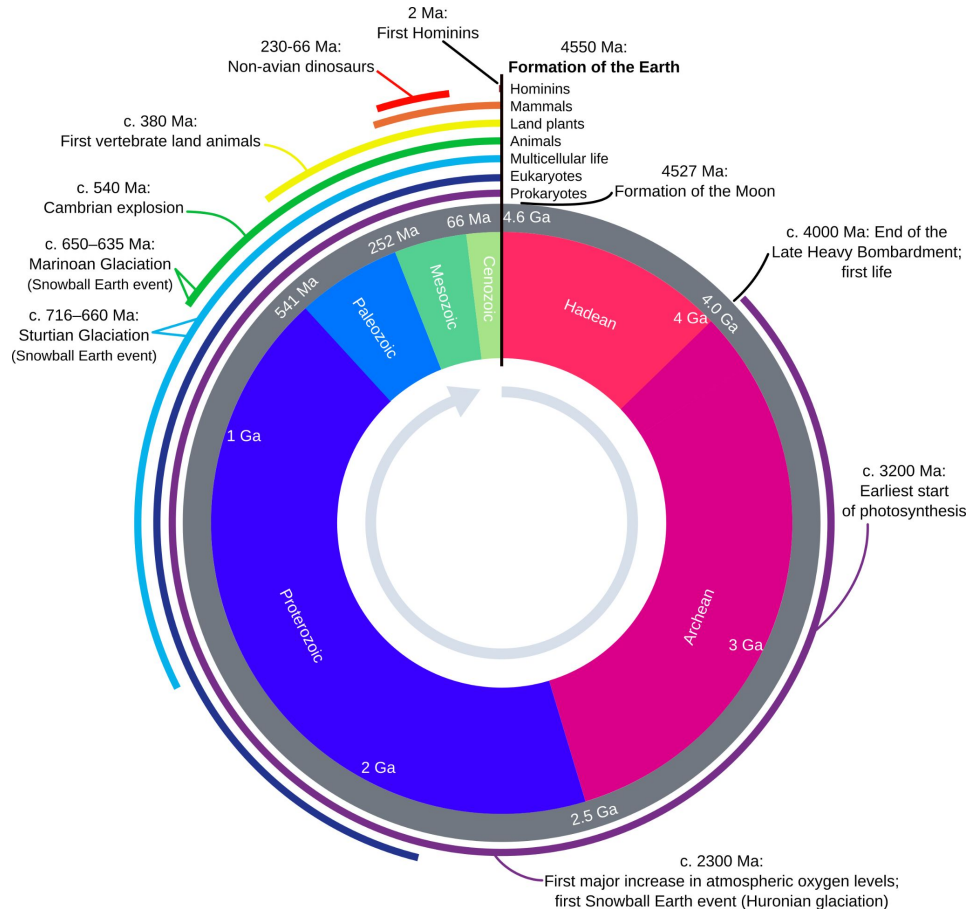


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Why Marine Microbes Matter ?

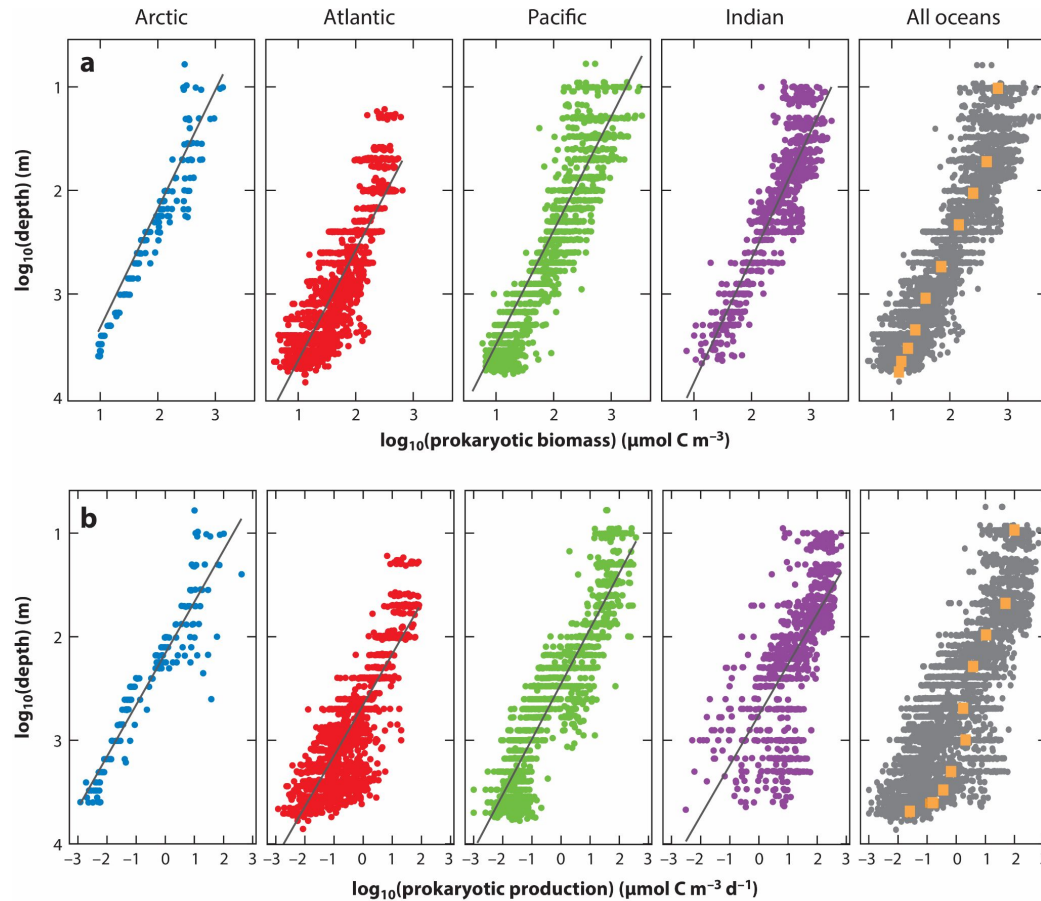


Origin and evolution of Life on Earth

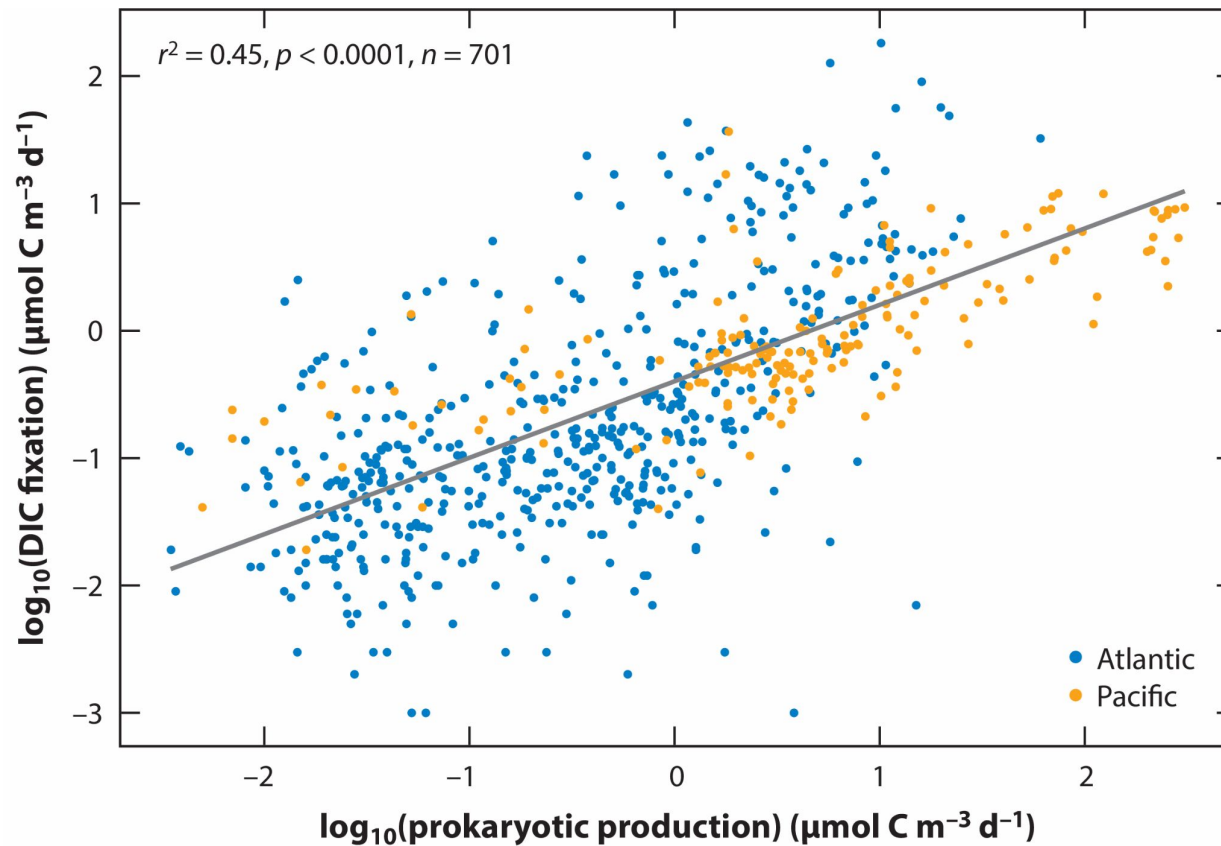


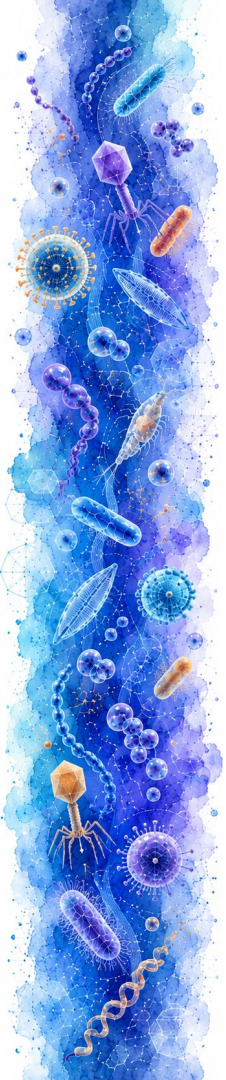
~45 % of Oxygen Production
~45 % of Primary Productivity
~90% of OM Remineralization
~100% of Nitrogen Fixation
~80% of Methane Emissions
... and much more

Why Marine Microbes Matter ?



Why Marine Microbes Matter ?





Basic Life Principles

Cellular Structure and Organization:

Minimal structural requirements for a functioning living system

Metabolic Diversity:

Diverse fuel and carbon sources (i.e., reduced/oxidized compounds, light, organic and inorganic carbon) show off the foundational energetic pathways that power microbial metabolisms

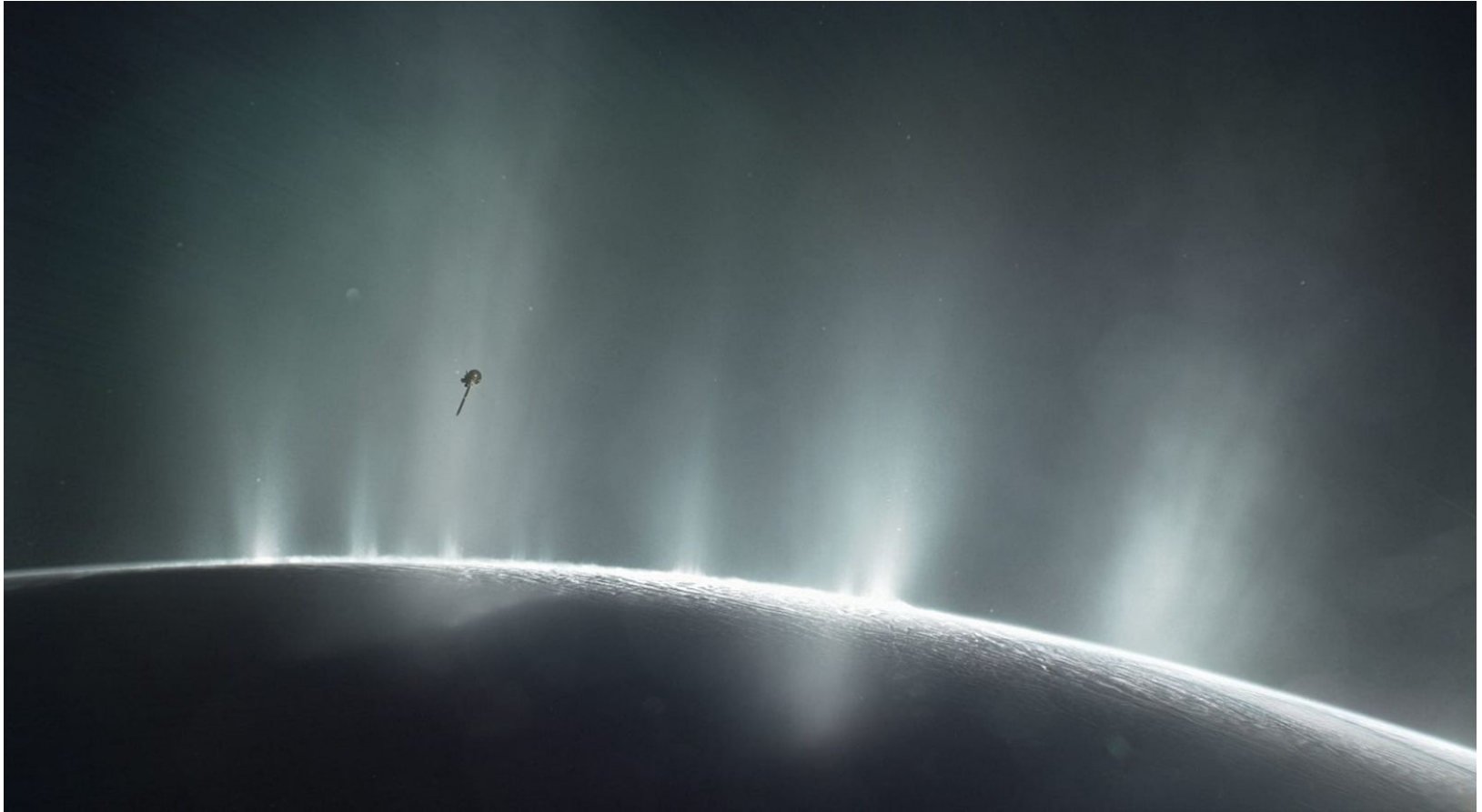
Information Flow and Genetics

Genetic data are stored and replicated via DNA and RNA. Rapid adaptation are carried out through mutation and horizontal gene transfer

Homeostasis and Growth

A balanced internal responses to external variables like temperature, pH, and nutrient availability, control microbial growth, mirroring how complex organisms maintain stability

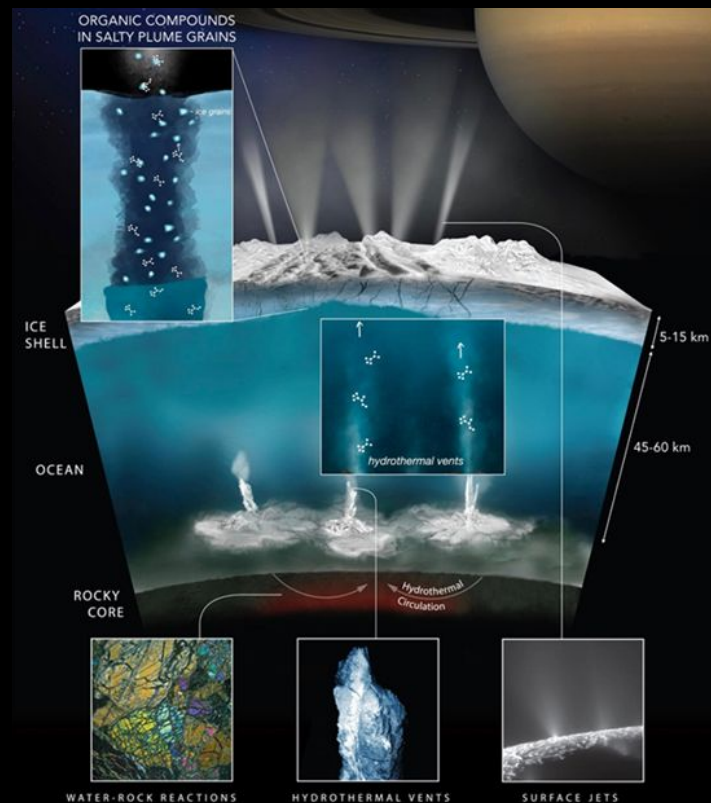
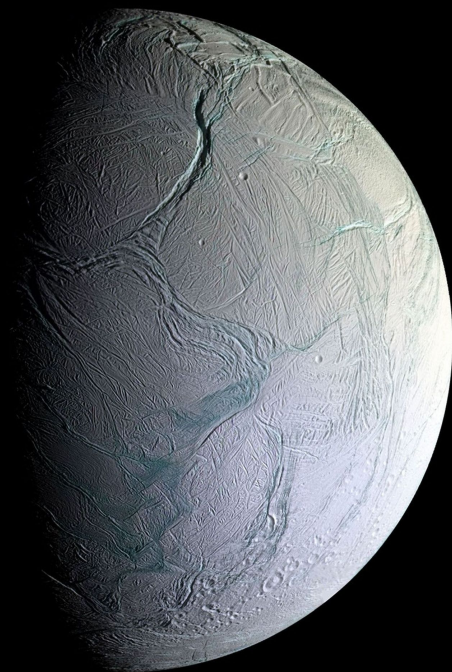
Extraterrestrial Life Models



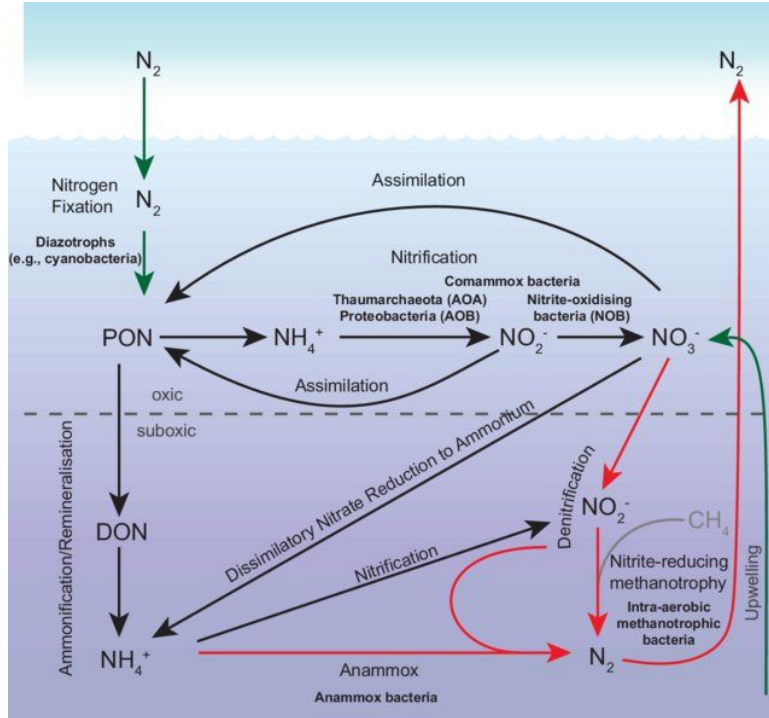
Credits: Cassini Grand Finale movie NASA / JPL-Caltech

Extraterrestrial Life Models

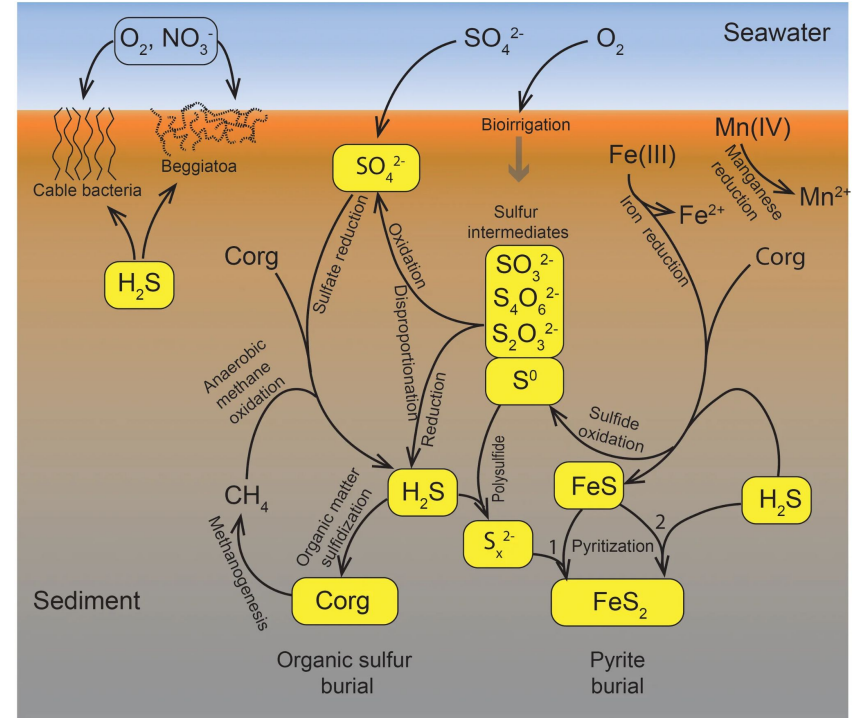
Saturn Icy Moon: Enceladus



Contribution To The Planet Functioning

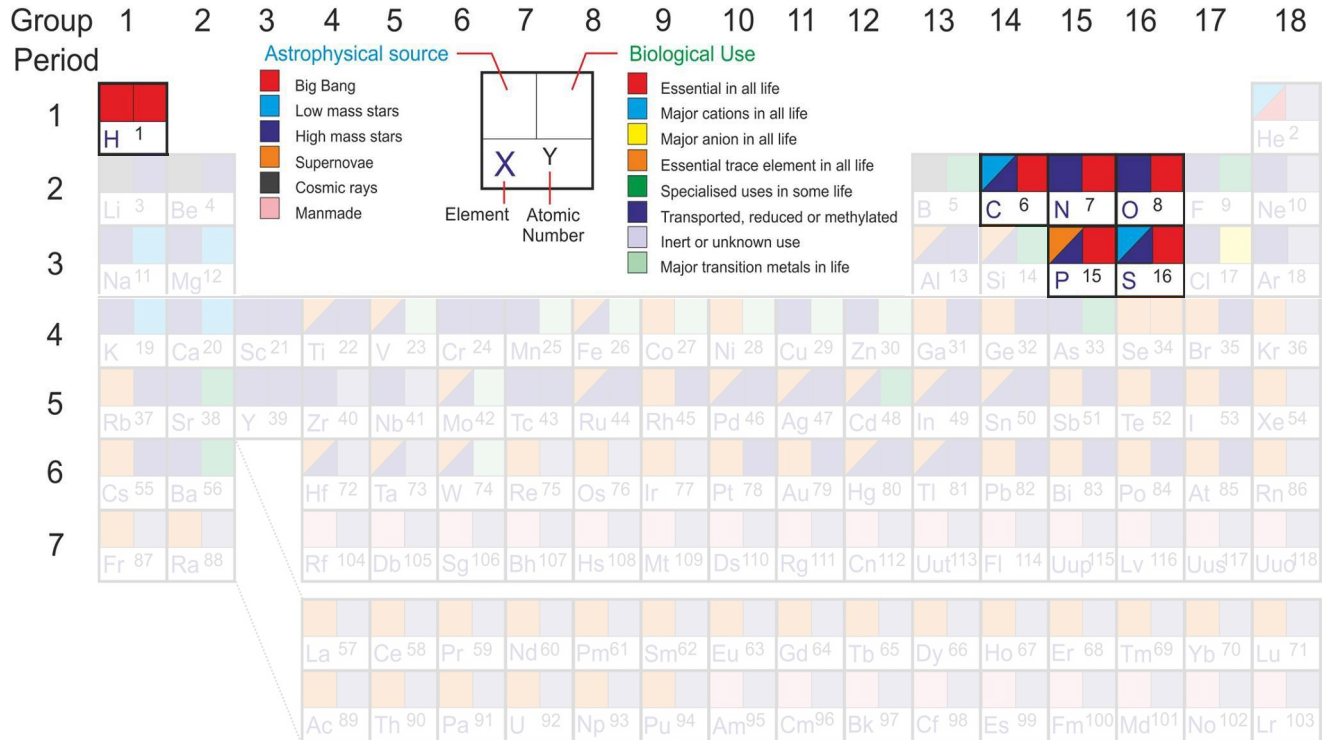


doi:10.1111/1462-2920.13682



<https://doi.org/10.3389/fmicb.2019.00849>

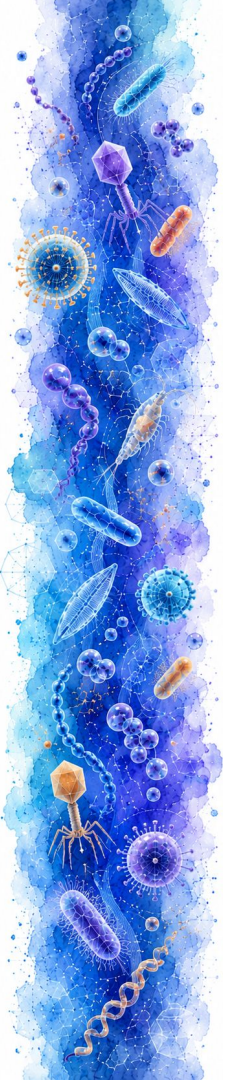
Contribution To The Planet Functioning



Biological data from Wackett, L.P., Dodge, A.G., Ellis, L.B.M. (2004) *Applied and Environmental Microbiology* **70**, 647-655.

A vertical illustration featuring a variety of microorganisms. At the top, there's a purple, multi-segmented worm-like structure. Below it, a blue, oval-shaped virus with a textured surface. To the right, a purple, multi-segmented worm-like structure. In the center, a blue, oval-shaped virus with a textured surface. To the right, a purple, multi-segmented worm-like structure. At the bottom, a blue, oval-shaped virus with a textured surface. The background is a mix of blue and purple watercolor-like textures with small white dots.

Biological data from Wackett, L.P., Dodge, A.G., Ellis, L.B.M. (2004) *Applied and Environmental Microbiology* **70**, 647-655.



Biotechnological Applications

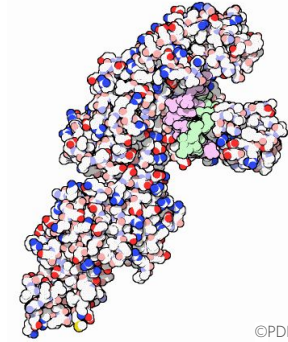
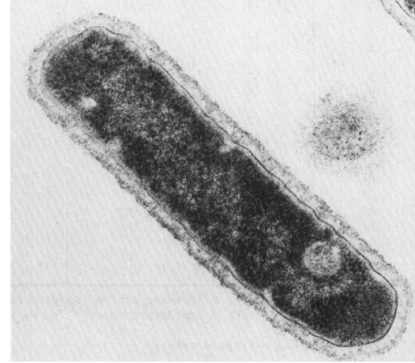
"Bacteria are able to grow [...] at any temperature at which there is liquid water, even in pools which are above the boiling point."

Thomas Brock

Biotechnological Applications

"Bacteria are able to grow [...] at any temperature at which there is liquid water, even in pools which are above the boiling point."

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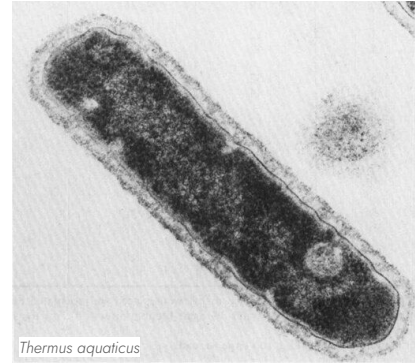


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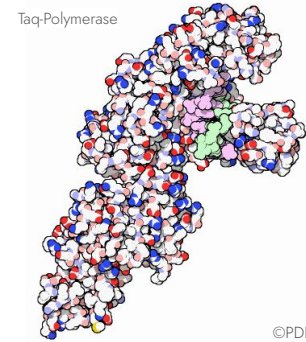
Biotechnological Applications

"Bacteria are able to grow [...] at any temperature at which there is liquid water, even in pools which are above the boiling point."

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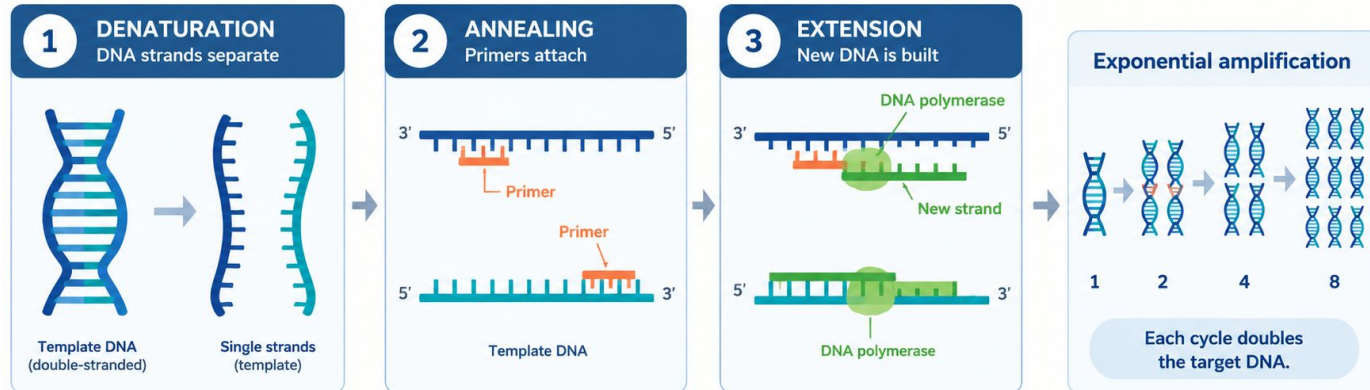


Thermus aquaticus



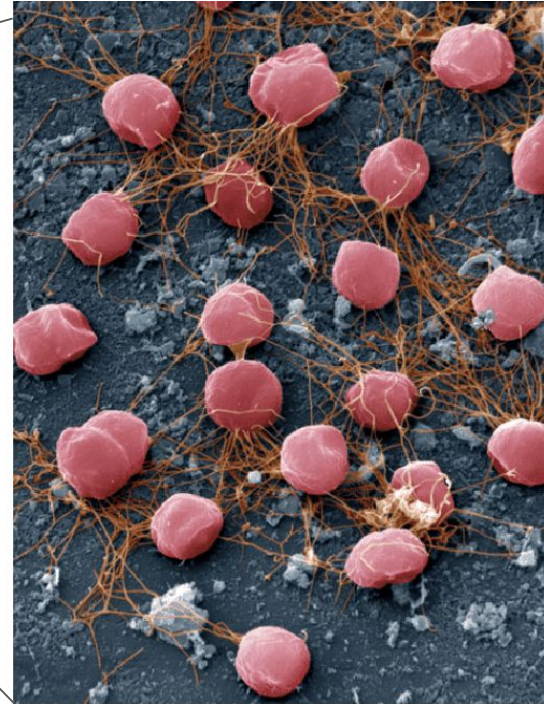
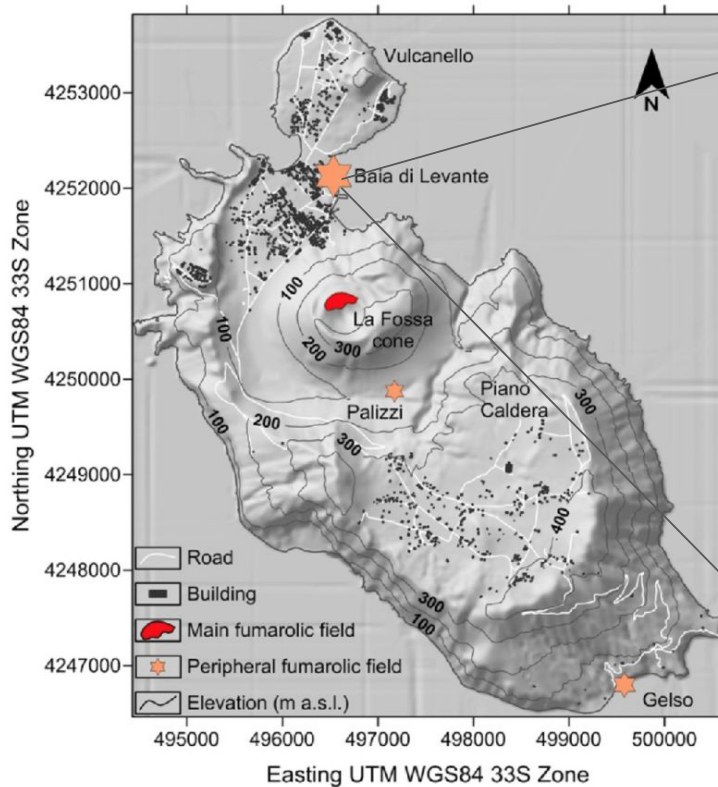
Taq-Polymerase

©PDB-101



Biotechnological Applications

Pyrococcus furiosus, a marine heterotrophic archaeebacteria growing optimally at 100 °C



Basic research

**Applied research
with sensational
findings**

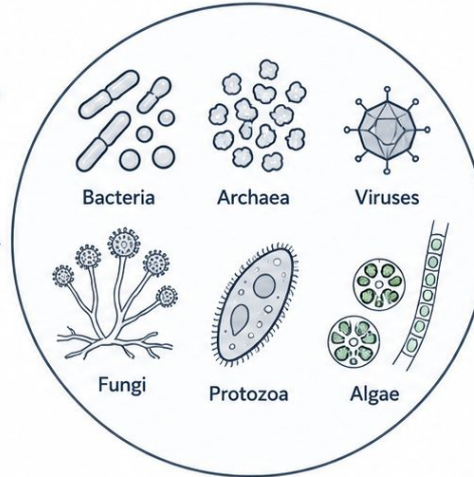
**Funding and
attention**

MICROBIOLOGY

μῖκρος — small

βίος — life

λόγος — speech



The scientific study of
microscopic organisms
and the microbial world.



BASIC SCIENCE

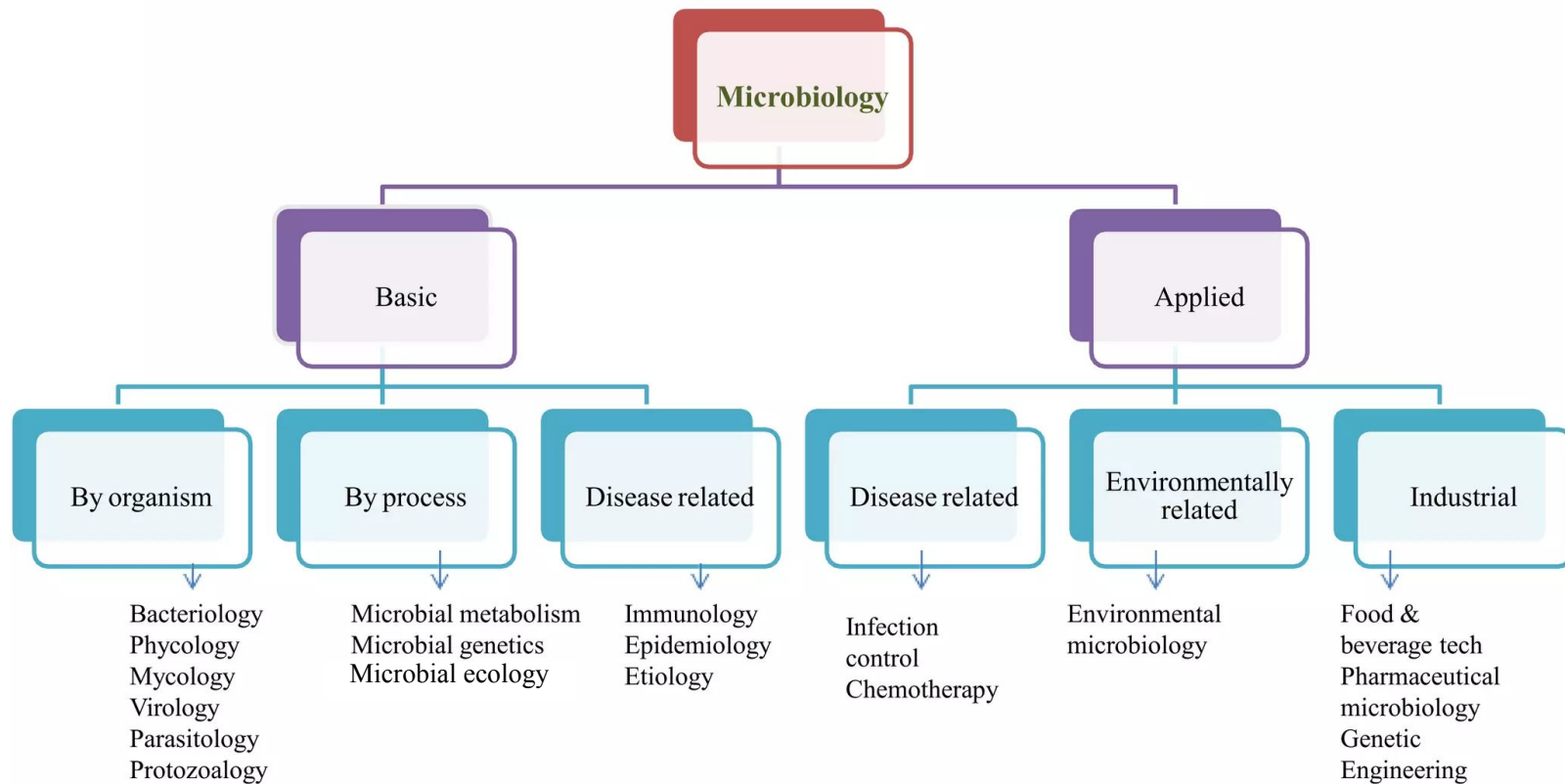
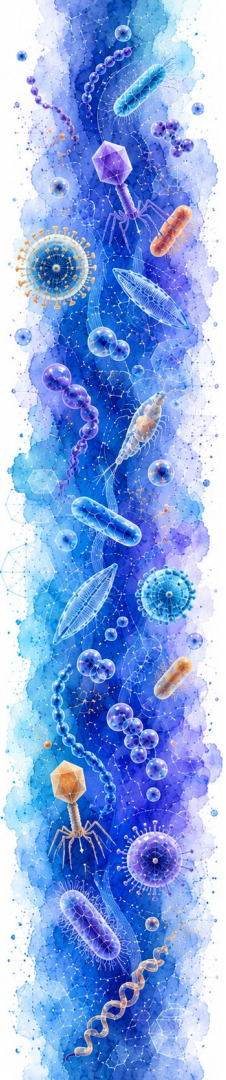
Understands the nature and
functioning of the microbial world.

Why? What? How?

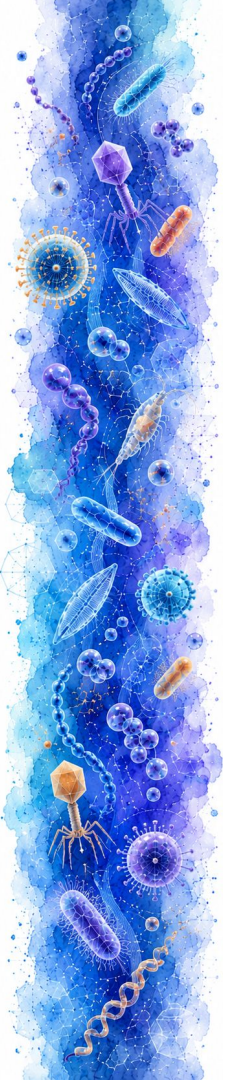
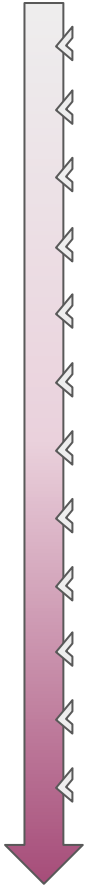


APPLIED SCIENCE

Uses knowledge of the microbial
world to solve practical problems
for the benefit of humankind
and planet Earth.



Brief History Of Microbiology

- 
- 
- 1665 – Hooke – Discovery of microorganisms
 - 1676 – Van Leeuwenhoek – First bacteria observation
 - 1768 – Spallanzani – First attempt of disproving spontaneous generation
 - 1866 – Cohn – Beggiatoa sulfur granules
 - 1876 – Koch – Koch's postulates
 - 1880 – Pasteur – Disproved spontaneous generation, the germ theory and first vaccine
 - 1884 – Gram – Gram staining technique
 - 1887 – Winogradsky – Chemolithoautotrophy
 - 1916 – Vernadsky – The biosphere concept
 - 1929 – Fleming – Penicillin
 - 1977 – Woese – Discover the Archaea
 - 1979 – Jannasch – Chemosynthesis sustains hydrothermal vent ecosystems

Prokaryotes



Bacteria: medical and environmental relevance



Not all bacteria are bad!



About >16,000 bacterial species are known today
(>150,000 if we include uncultured).



538 are **known pathogens** or **facultative pathogens** (~0.36%)



Prokaryotes



Prokaryotes include the ***Bacteria*** and the ***Archaea*** and consist of small and structurally rather simple cells.

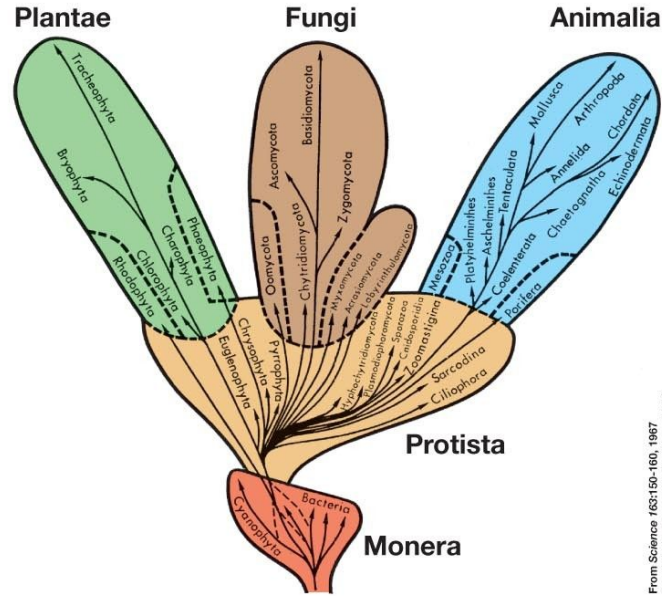
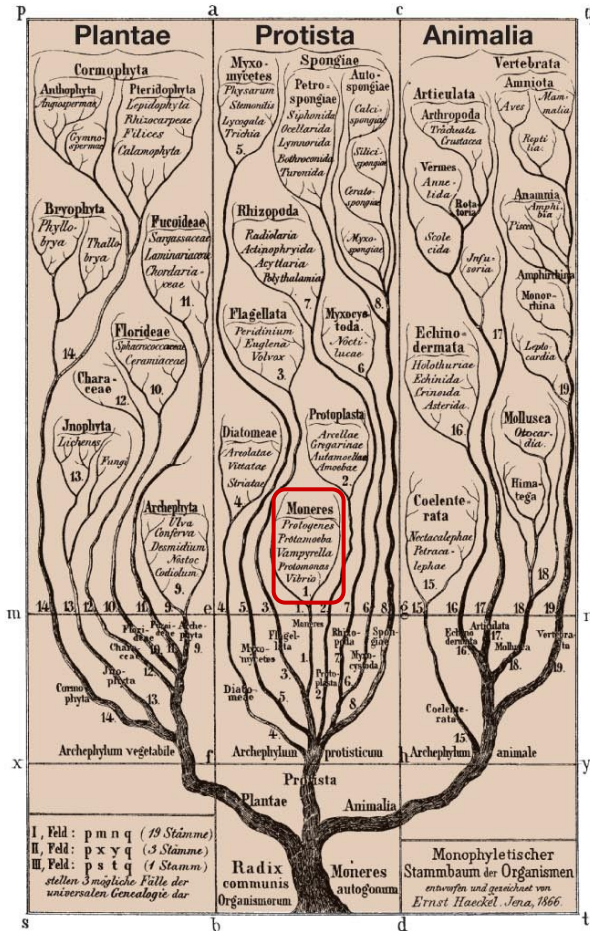


The word “Prokaryote” does not imply evolutionary relatedness.



Bacteria and ***Archaea*** are two distinct domain of life, similarly to *Bacteria* and *Eukarya*.

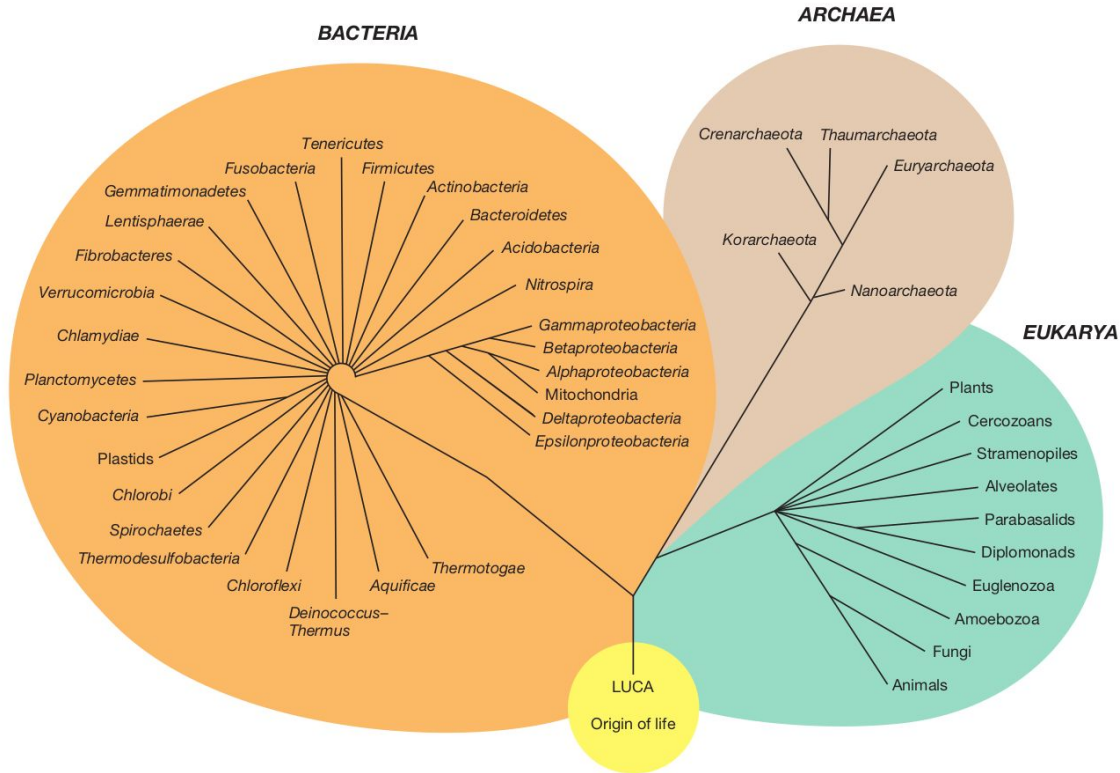




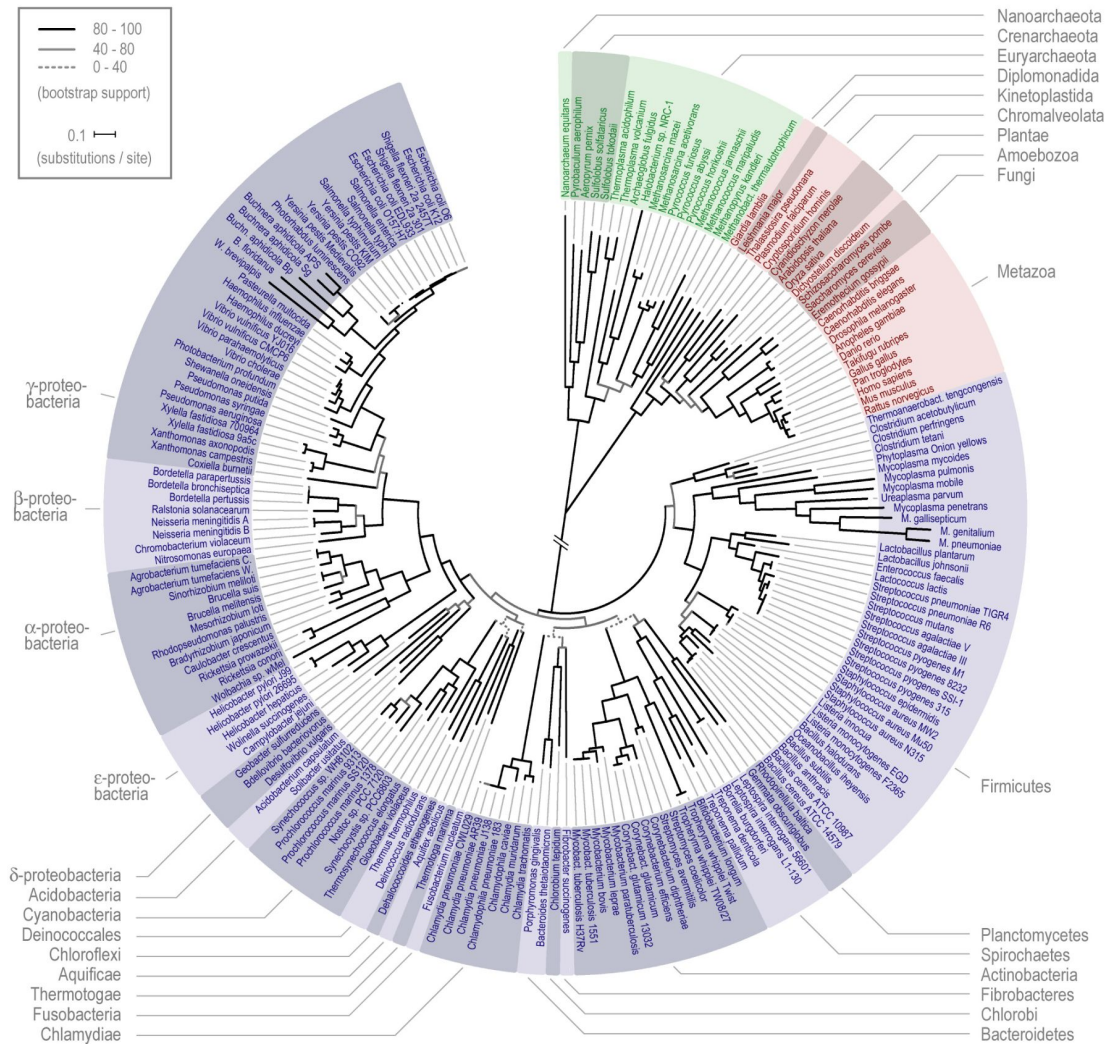
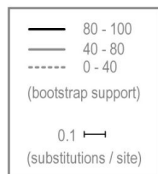
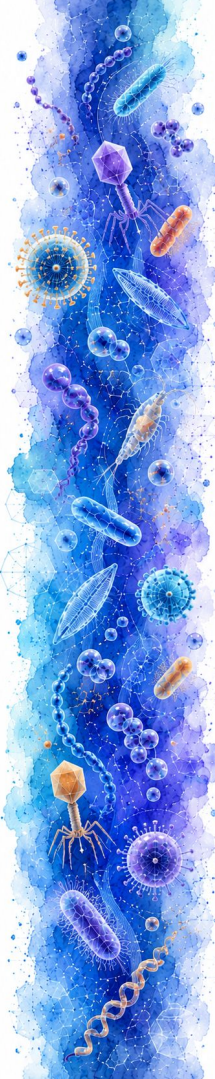
From *Science* 163:150-160, 1967
Reprinted with permission from AAAS

Monere lead the way to Prokaryotes which became *Eubacteria* and *Archaeobacteria* before becoming *Bacteria* and *Archaea*

The universal tree of life based on rRNA gene sequences



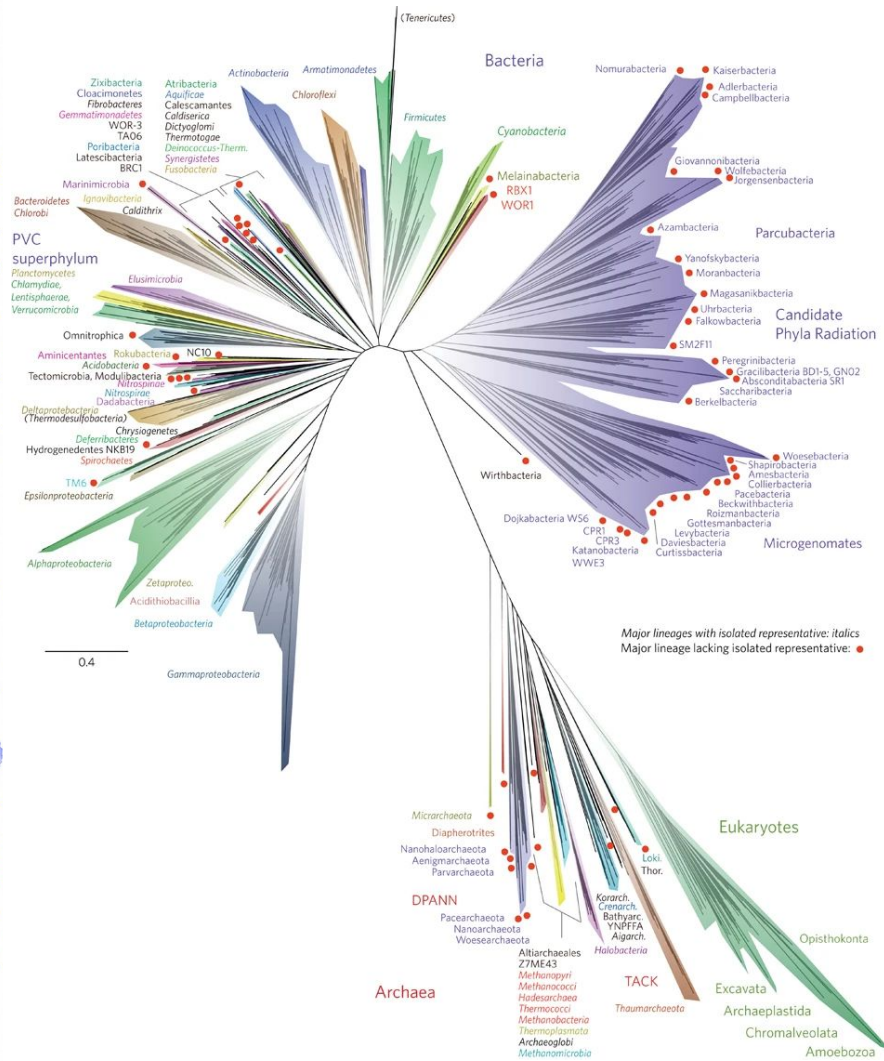
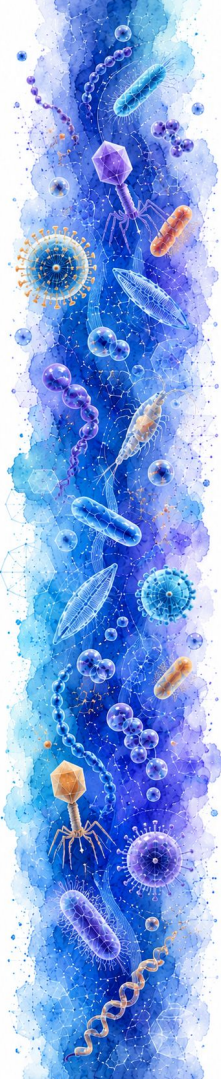
...a phylogenetic tree that depicts the evolutionary history of all cells and clearly reveals the three domains



2006

ARCHAEA
 EUKARYA
 BACTERIA

Global phylogeny of fully sequenced organisms. The phylogenetic tree has its basis in a cleaned and concatenated alignment of 31 universally occurring genes coding for universal protein families, with indisputable orthology in 191 species with completely annotated genome



2016

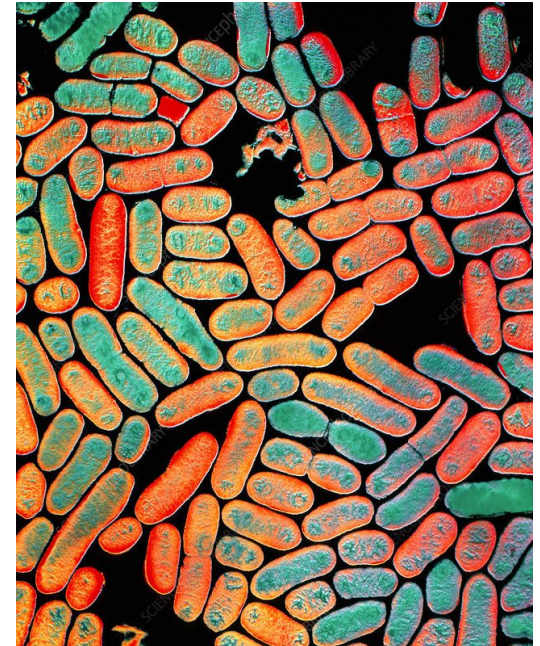
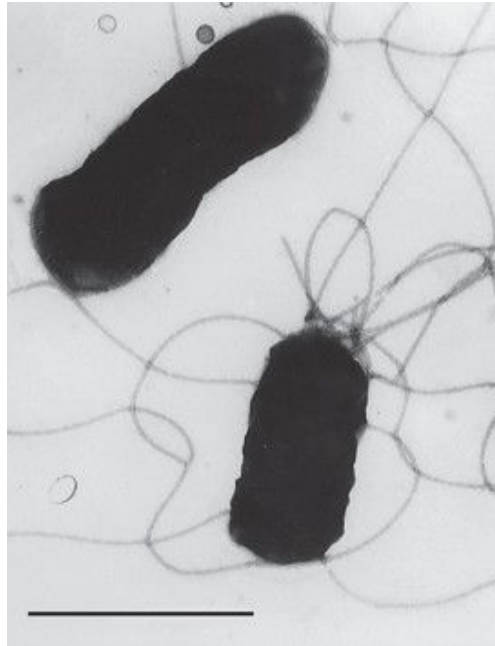
ARCHAEA
EUKARYA
BACTERIA

A new view of the tree of life. Based on ribosomal protein sequences, the tree includes 92 named bacterial phyla, 26 archaeal phyla and all five of the Eukaryotic supergroups PLUS new genomic data from over 1,000 uncultivated and little known organisms, together with published sequences, to infer a dramatically expanded version of the tree of life

Microbial Species Concept: Taxonomy

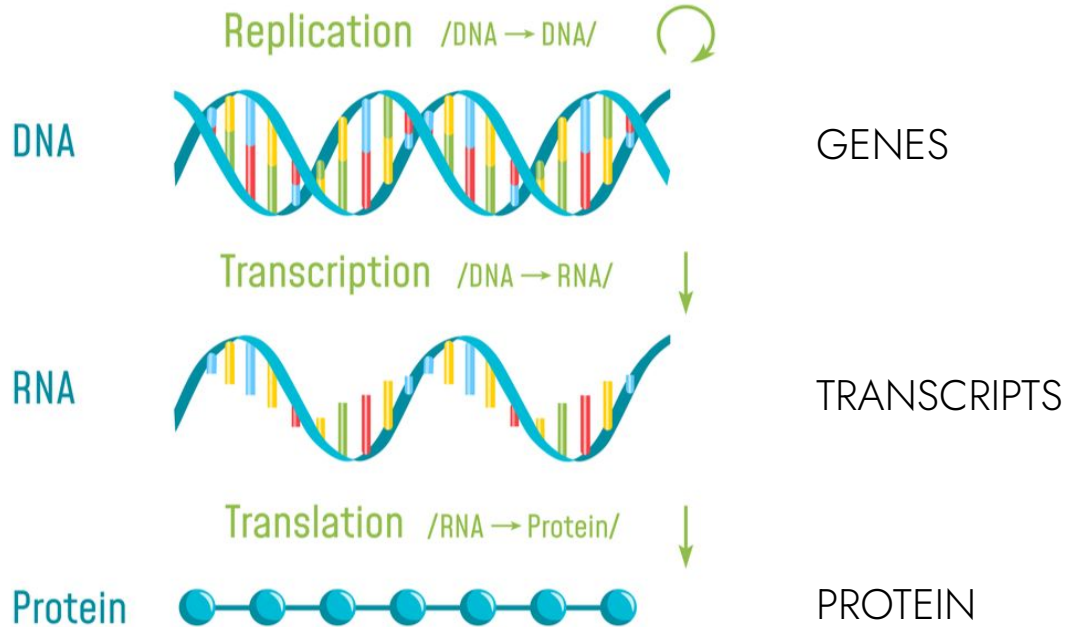
Taxonomic nomenclature follows the binomial systems. There are numerous rules and conventions, all established by the *International Committee on the Systematics of Prokaryotes*.

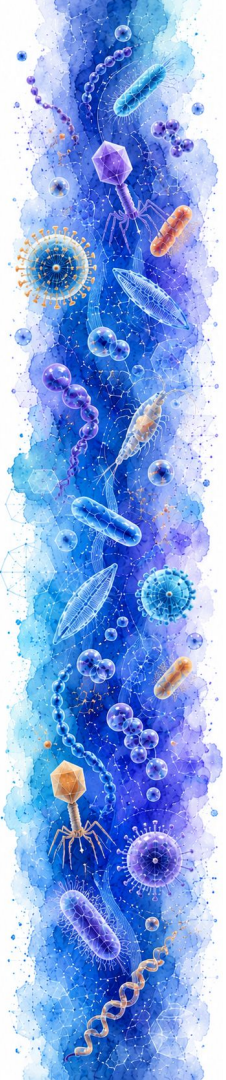
Domain: Bacteria
Phylum: Pseudomonadota
Class: Gammaproteobacteria
Order: Enterobacteriales
Family: Enterobacteriaceae
Genus: Escherichia
Species: Escherichia coli
Strain: K12



Genes And Genomes

Central Dogma of Molecular Biology





Genes And Genomes

Central Dogma of Molecular Biology - Is Still Valid ?

- .**Alternative Splicing:** A single gene can be cut and pasted in different ways during the mRNA stage. This allows one gene to code for multiple distinct proteins, expanding cellular complexity
- .**Non-Coding RNAs:** Many genes copy their DNA into RNA, but that RNA never gets translated into a protein. Instead, molecules like microRNA (miRNA) or long non-coding RNA (lncRNA) act as regulatory switches inside the cell
- .**Reverse Transcription:** Some entities, like retroviruses (e.g., HIV), can reverse the direction of information, copying RNA back into DNA using an enzyme called reverse transcriptase

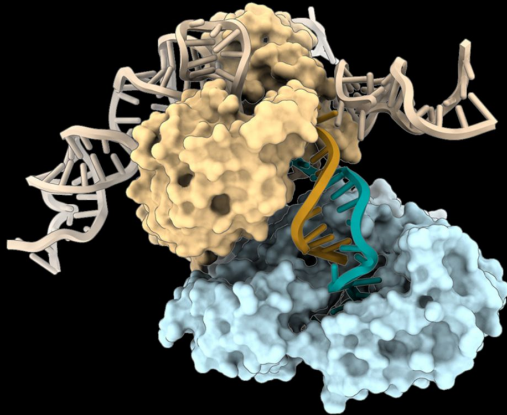
Genes And Genomes

Central Dogma of Molecular Biology - Is Still Valid ?

Scientists stunned by 'fundamentally new way' life produces DNA

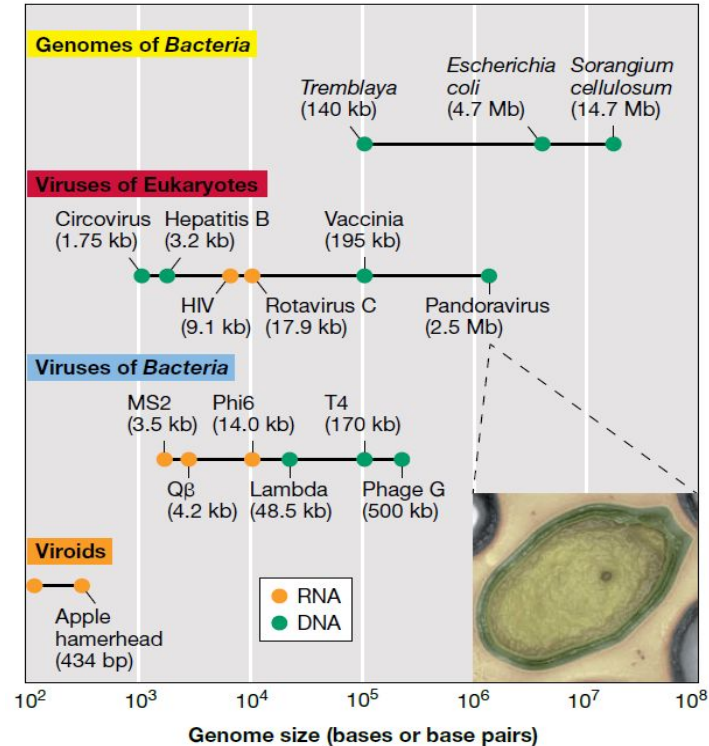
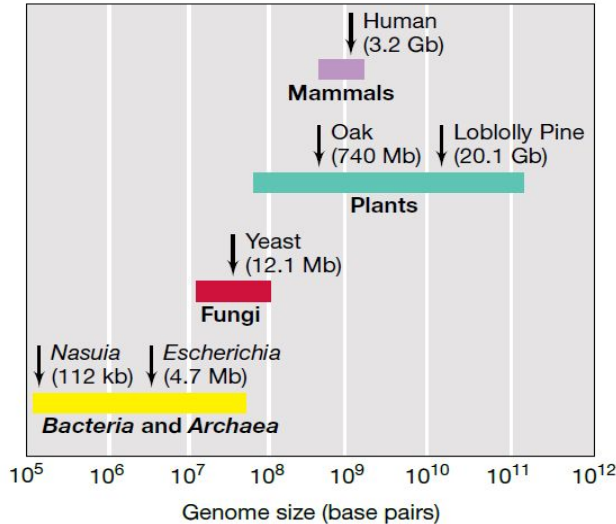
Bacterial system uses protein in novel way to build mysterious repetitive DNA sequence that defends against viruses

18 APR 2016 2:00 PM ET BY RICHARD STONE



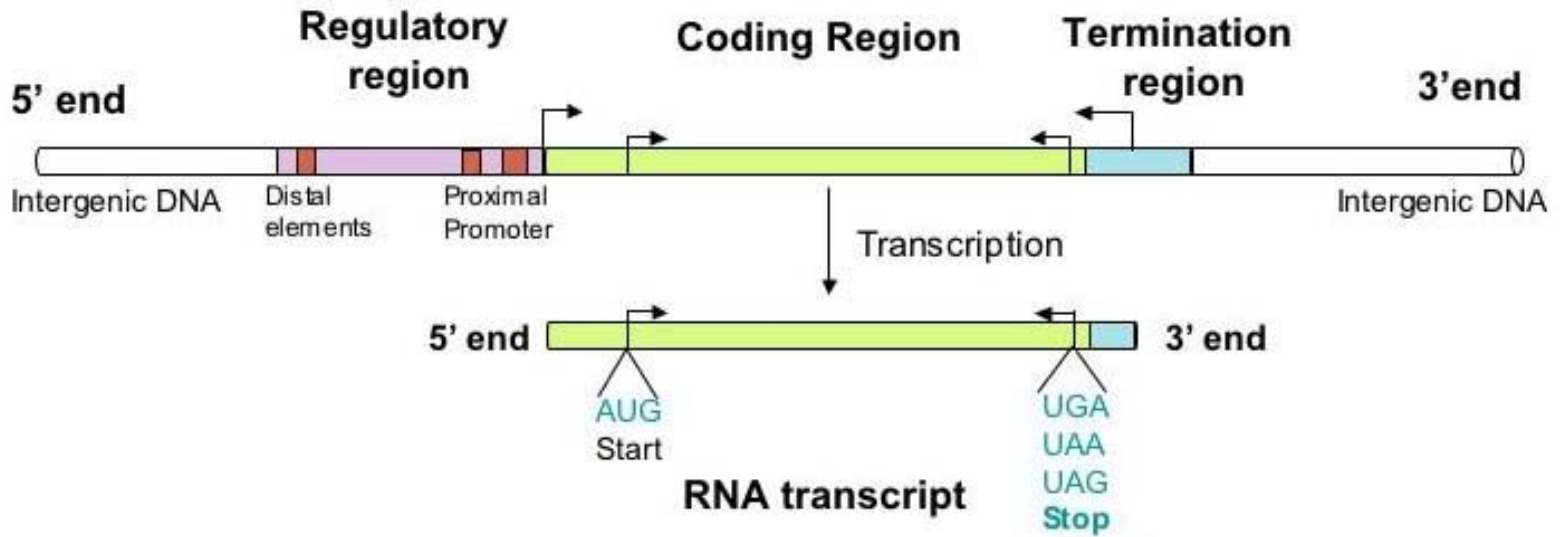
A Stanford University team describes a **bacterial enzyme that synthesizes a long repetitive DNA sequence without a nucleic acid template, using its own structure as a guide.** The new finding centers on DRT3, a defense system that **protects bacteria from viruses**, known as phages, that infect them. DRT3, the researchers found, bypasses the logic of base pairing. It relies on two reverse transcriptases: a conventional one that builds single-stranded DNA from an RNA template, and a second, unusual one that assembles its complement from its own built-in template. This unusual enzyme, called Drt3b, has amino acids in its active site that mimic a template RNA strand.

Genes And Genomes

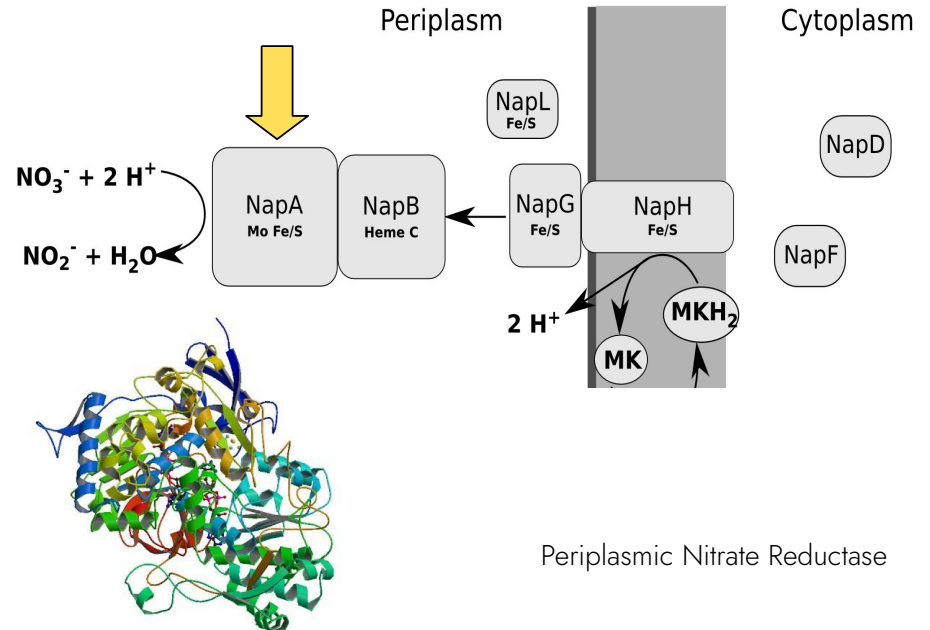
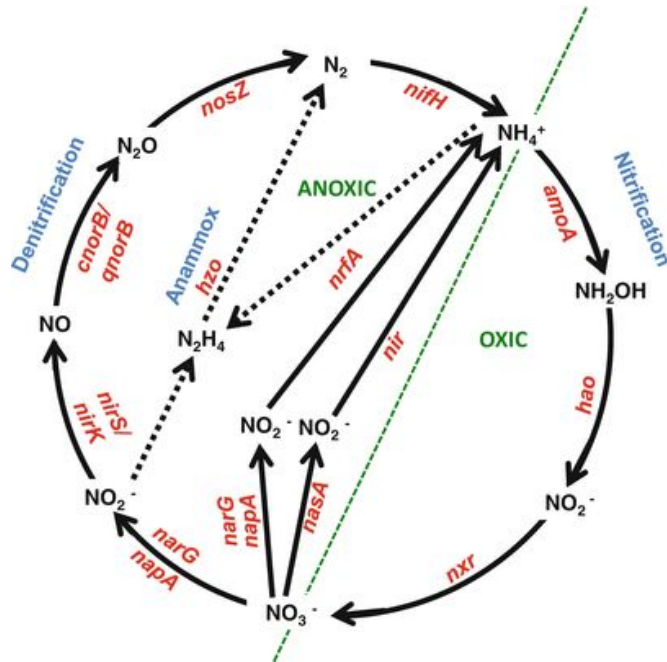
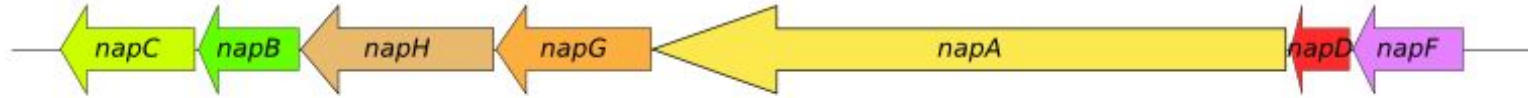


Prokaryotic genomes range in size from those of large viruses to those of eukaryotic microbes. Prokaryotic genomes are generally circular. One or more chromosome may be present. One or more plasmid can be present (circular or linear), coding (often) for non essential genes conferring additional characteristics.

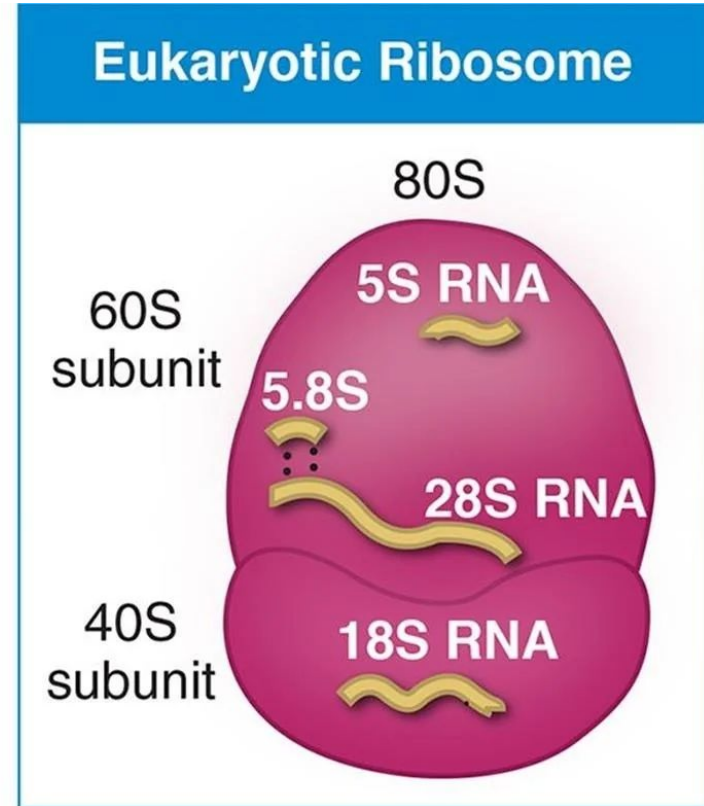
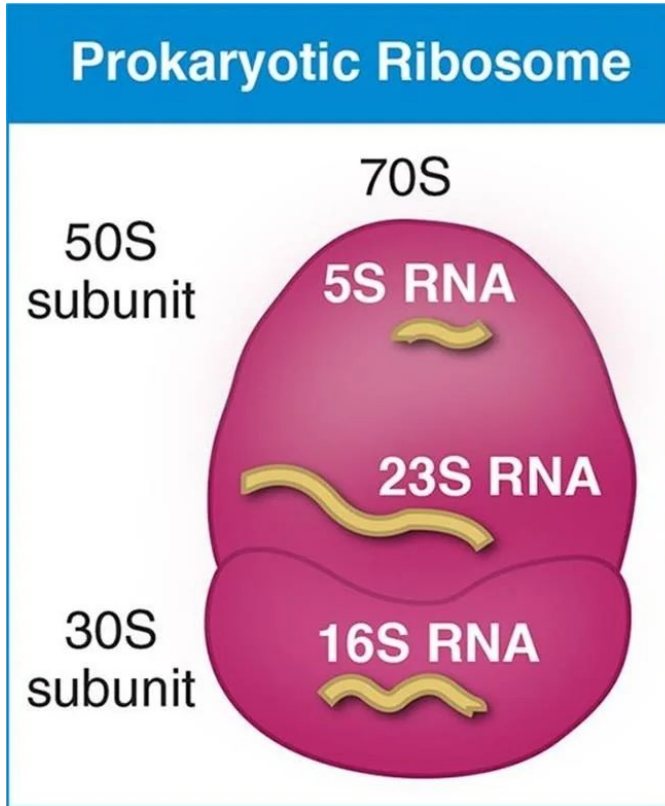
Gene Organization In Prokaryotes



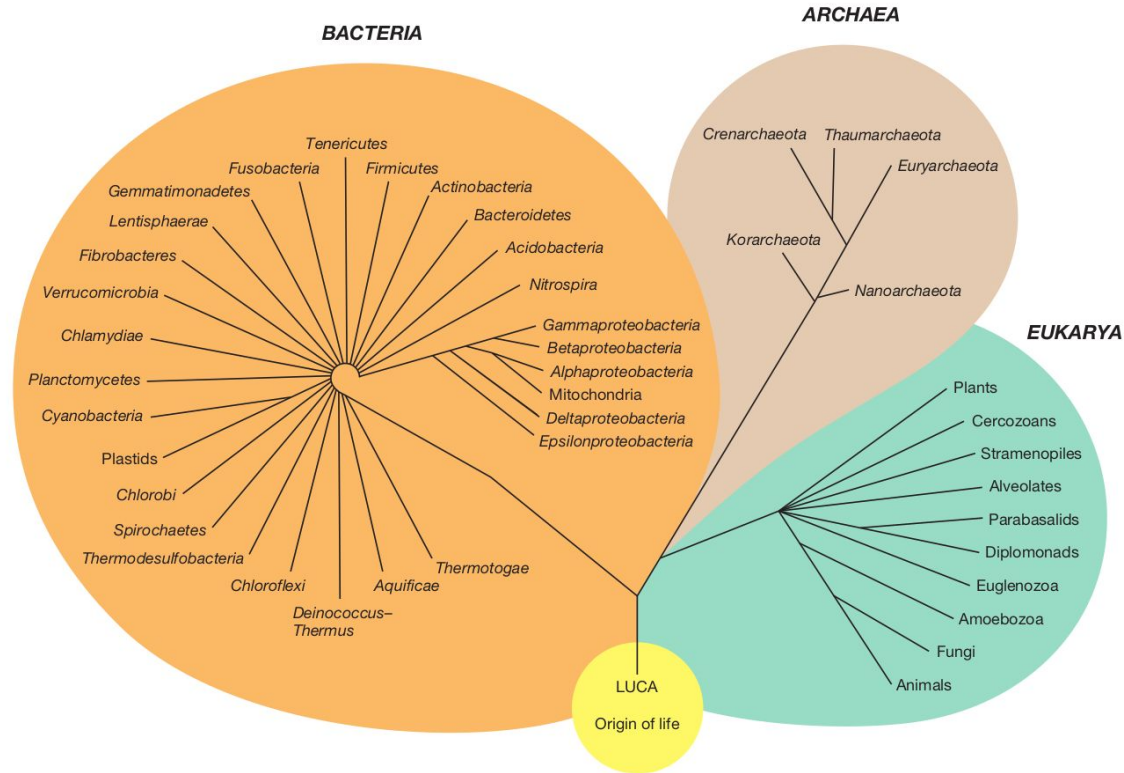
Visualizing Genes Organization



Diversity And Marker Genes



The universal tree of life based on rRNA gene sequences



...a phylogenetic tree that depicts the evolutionary history of all cells and clearly reveals the three domains

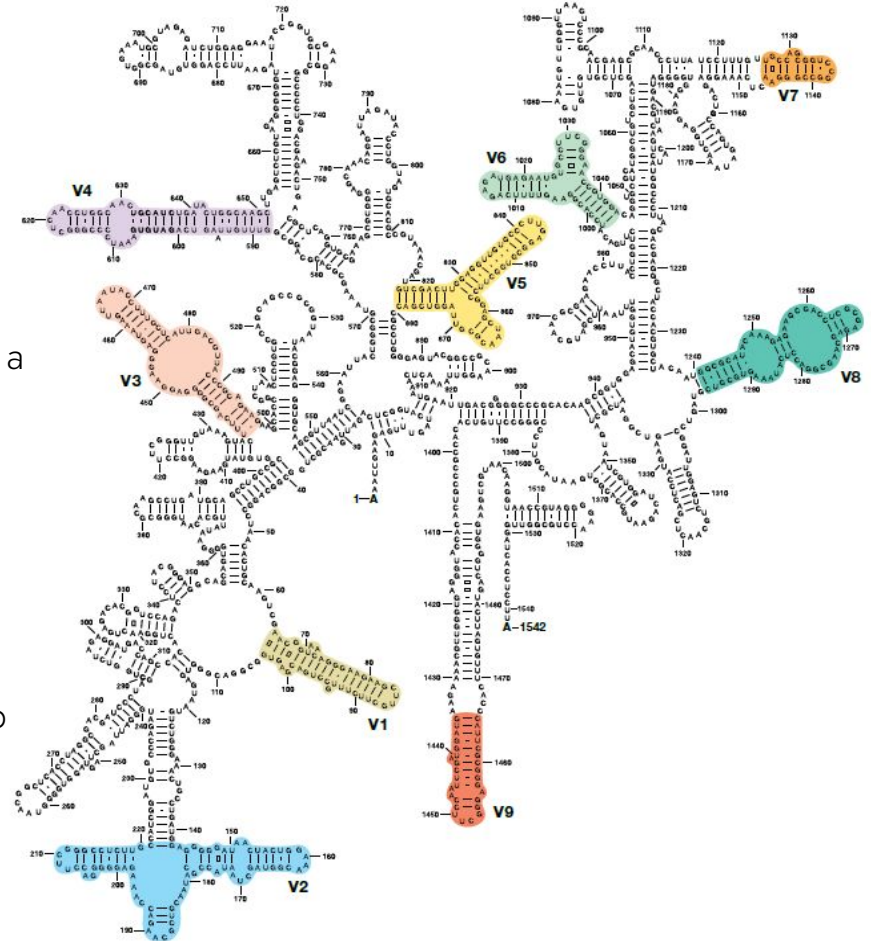
Diversity And Marker Genes

16S ribosomal RNA (or 16S rRNA) is the RNA component of the 30S subunit of a prokaryotic ribosome (SSU rRNA).

The 16S rRNA gene is the most widely used marker gene to investigate prokaryotic diversity in a sample and to compare it across samples.

In addition to highly conserved primer binding sites, 16S rRNA gene sequences contain hypervariable regions that can provide species-specific signature sequences useful for identification of bacteria.

Different regions are used depending on sequencing technology, scientific question, group investigated.



Visualizing Genes Organization

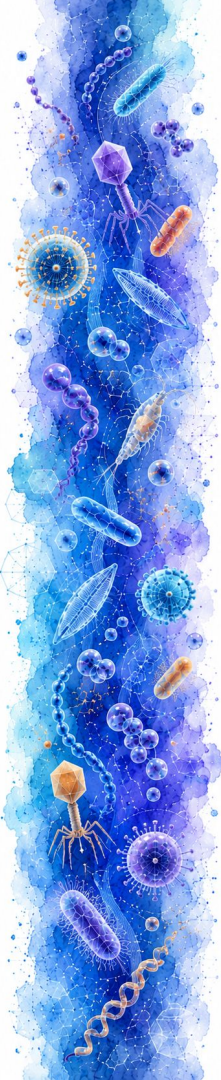
CURRENTLY AVAILABLE AMPLICON TARGETS/PRIMERS (RECOMMENDED SETS IN BOLD)

					Primer Set Coverages (SILVA TestPrime, 0-2 mismatches) ^a					
Primer Targets	Region(s)	Forward Primer	Reverse Primer	Source(s)	Archaea	Bacteria	Cyanos	Eukarya	mtDNA	chIDN/
Illumina MiSeq short variable region targets										
Standard rRNA targets										
Universal	V4-V5 ^b	515FB = GTGYCAGCMGCCGCGGTAA	926R = CCGYCAATTYMTTTRAGTTT	Parada 2015 / Walters 2015	81-93%	85-95%	85-94%	81-94%	57-77%	81-93%
Archaea-specific	16S V6-V8	A956F = TYAATYGGANTCAACRCC	A1401R = CRGTGWGTRCAAGGRGCA	Comeau 2011	71-82%	-	-	0-89%	0-1%	0-1%
Bacteria-specific	16S V6-V8	B969F = ACGCGHNRAACCTTACC	BA1406R = ACGGGCRGTGWGTRCAA	Comeau 2011	0-14%	72-83%	66-88%	0-1%	14-75%	47-87%
Bacteria-specific	16S V3-V4	341F = CCTACGGGNGGCWGCAG	805R = GACTACHVGGGTATCTAATCC	Illumina / Klindworth 2013	0-90%	83-95%	71-93%	-	11-54%	49-90%
Eukaryote-specific	18S V4	E572F = CYGCGGTAATCCAGCTC	E1009R = AYGGTATCTRATCRTCTTYG	Comeau 2011	-	-	-	54-92%	1%	1%
Fungi-specific	ITS2 ^c	ITS86(F) = GTGAATCATCGAATCTTTGAA	ITS4(R) = TCCTCCGCTTATTGATATGC	Op De Beeck 2014	n/a	n/a	n/a	n/a	n/a	n/a
Bacteria-specific	16S V1-V3 ^d	27Fmod = AGRGTTTGATCMTGGCTCAG	519R = GWATTACCGCGGCKGCTG	Kim 2013 / Lane 1985	-	73-93%	44-89%	-	9-75%	24-87%
Bacteria+Archaea-specific ("EMP")	16S V4 ^e	515FB = GTGYCAGCMGCCGCGGTAA	806RB = GGACTACNVGGGTWTCTAAT	Walters 2015	84-96%	84-95%	76-92%	0-19%	52-89%	64-89%
Cyano-specific	16S V3-V4 ^f	CYA359F = GGGGAATYTTCCGCAATGGG	CYA781R = GACTACWGGGGTATCTAATCCCWTT	Nübel 1997	-	2-5%	58-88%	-	1-3%	35-79%

Visualizing Genes Organization

CURRENTLY AVAILABLE AMPLICON TARGETS/PRIMERS (RECOMMENDED SETS IN BOLD)

					Primer Set Coverages (SILVA TestPrime, 0-2 mismatches) ^a					
Primer Targets	Region(s)	Forward Primer	Reverse Primer	Source(s)	Archaea	Bacteria	Cyanos	Eukarya	mtDNA	chIDN/
Illumina MiSeq short variable region targets										
Standard rRNA targets										
Universal	V4-V5 ^b	515FB = GTGYCAGCMGCCGCGGTAA	926R = CCGYCAATTYMTTTRAGTTT	Parada 2015 / Walters 2015	81-93%	85-95%	85-94%	81-94%	57-77%	81-93%
Archaea-specific	16S V6-V8	A956F = TYAATYGGANTCAACRCC	A1401R = CRGTGWGTRCAAGGRGCA	Comeau 2011	71-82%	-	-	0-89%	0-1%	0-1%
Bacteria-specific	16S V6-V8	B969F = ACGCGHNRAACCTTACC	BA1406R = ACGGGCRGTGWGTRCAA	Comeau 2011	0-14%	72-83%	66-88%	0-1%	14-75%	47-87%
Bacteria-specific	16S V3-V4	341F = CCTACGGGNGGCWGCAG	805R = GACTACHVGGGTATCTAATCC	Illumina / Klindworth 2013	0-90%	83-95%	71-93%	-	11-54%	49-90%
Eukaryote-specific	18S V4	E572F = CYGCGGTAATCCAGCTC	E1009R = AYGGTATCTRATCRTCTTYG	Comeau 2011	-	-	-	54-92%	1%	1%
Fungi-specific	ITS ^{2c}	ITS86(F) = GTGAATCATCGAATCTTTGAA	ITS4(R) = TCCTCCGCTTATTGATATGC	Op De Beeck 2014	n/a	n/a	n/a	n/a	n/a	n/a
Bacteria-specific	16S V1-V3 ^d	27Fmod = AGRGTTTGATCMTGGCTCAG	519R = GWATTACCGCGGCKGCTG	Kim 2013 / Lane 1985	-	73-93%	44-89%	-	9-75%	24-87%
Bacteria+Archaea-specific ("EMP")	16S V4 ^e	515FB = GTGYCAGCMGCCGCGGTAA	806RB = GGACTACNVGGGTWTCTAAT	Walters 2015	84-96%	84-95%	76-92%	0-19%	52-89%	64-89%
Cyano-specific	16S V3-V4 ^f	CYA359F = GGGGAATYTTCCGCAATGGG	CYA781R = GACTACWGGGTATCTAATCCCWTT	Nübel 1997	-	2-5%	58-88%	-	1-3%	35-79%



Why 16S rRNA Became A Standard Marker For Bacterial Identification

.Universally distributed

Crucial component for protein synthesis (**survival**)



Detect unknown, uncultured, or rare bacteria in a sample without prior knowledge of what is present

.Conserved and Variable Regions

Conserved regions, stable across diverse bacterial phyla, useful universal PCR primers designing to bind and amplify the gene in almost any bacterium

Hypervariable regions (**V1–V9**) accumulate differences between species, functioning as a molecular fingerprint to differentiate taxonomic groups

.Low Rate of Mutation

Evolves slowly because changes in vital functional areas can disrupt the ribosome and provides enough stability to accurately map broad phylogenetic relationships

.Minimal Horizontal Gene Transfer

Historically considered resistant to horizontal gene transfer (HGT), meaning the gene descends **vertically** and faithfully tracks core evolutionary history. However, HGT and intragenomic variation can **occasionally occur**, meaning 16S rRNA is highly reliable for **genus-level identification** but can struggle to cleanly separate closely related species

Horizontal Gene Transfer (HGT)

Mechanism of transfer of genetic information between organisms, as opposed to vertical inheritance from parental organism(s).

It is considered one of the major mode of prokaryotic evolution.

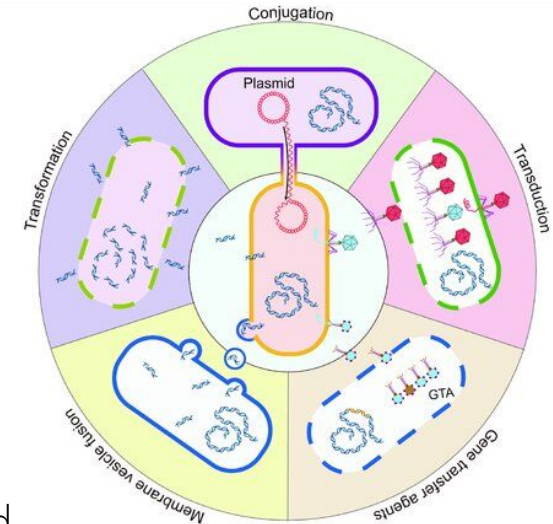
.Conjugation (direct contact): the DNA of the donor bacteria is transmitted to the recipient bacteria by the conjugative pili or adhesins

.Transformation: the lysed bacteria release naked DNA fragments, which are acquired, integrated, and expressed by other bacteria

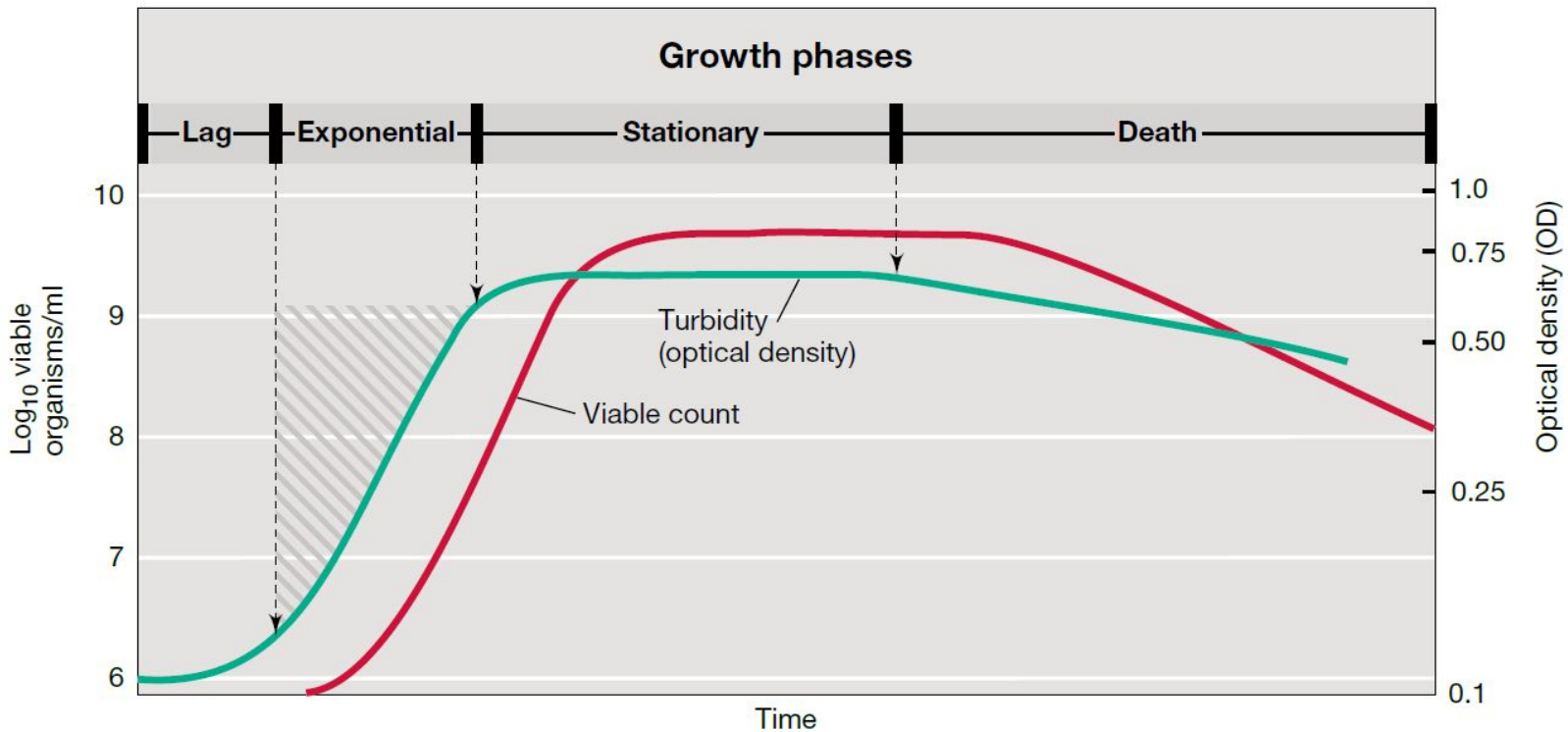
.Transduction: a type of HGT mediated by bacteriophage. After the bacteriophage infects the bacteria, bacterial DNA fragments might be accidentally loaded into the bacteriophage head. The bacteriophage that carries the host bacteria's DNA is released and infects a new bacterium to complete HGT.

.Membrane vesicle fusion (MVs): the bacterial outer membrane bulges to form 20-250 nm MVs that carry genetic material and releases it into the environment.

.Gene Transfer Agents (GTAs): phage-like particles that are produced by bacteria, which carry random DNA fragments from the original bacteria and transfer this into the recipient bacteria



Microbial growth



$$\text{OD}_{600} 1.0 = 8 \times 10^8 \text{ cells/ml}$$

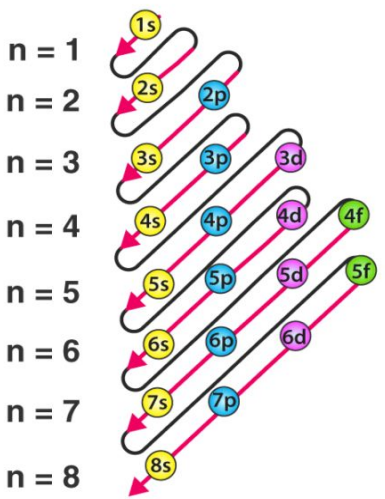
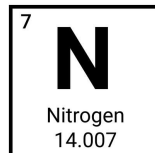
[illegible]University of Naples Federico II

Valence Electrons And Oxidation Numbers

Valence refers to the ability of an atom to form bonds with other atoms.

Generally, the **valence** of an atom is defined by the number of electrons in the outermost shell (stated by the electron configuration) it can lose, gain, or share.

Oxidation numbers (or oxidation states) are hypothetical charges assigned to atoms within a molecule or ion, and correspond to the number of electrons effectively participating in redox reactions.

		Number of electrons	Electron configuration					Valence electrons
		7						5
			Oxidation numbers					
	-3	0	+1	+2	+3	+4	+5	
								
	Ammonia/ Ammonium	Dinitrogen	Dinitrogen oxide	Nitrogen Oxide	Nitrous Acid/ Nitrite	Nitrogen dioxide	Nitric Acid/ Nitrate	

Redox

Reducing Agent



A loses electrons

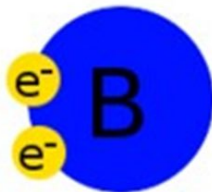


A is oxidized

Oxidizing Agent



B gains electrons



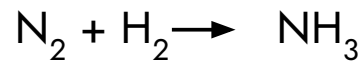
B is reduced

OXIDATION

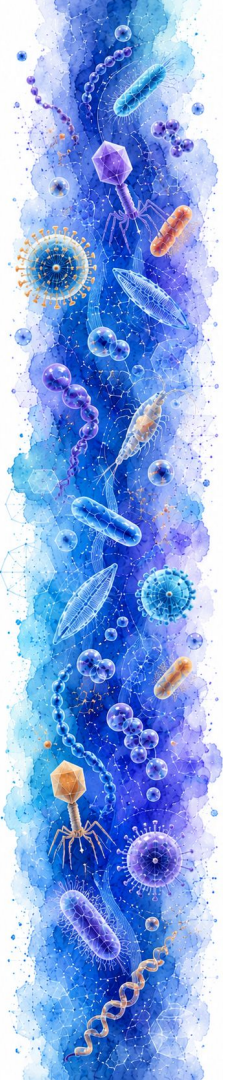


REDUCTION

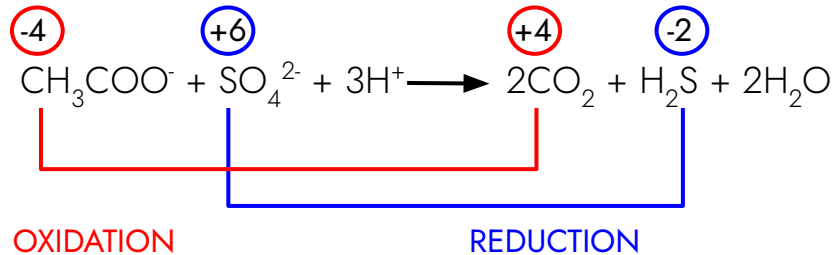
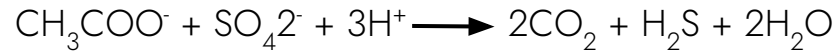
OXIDATION

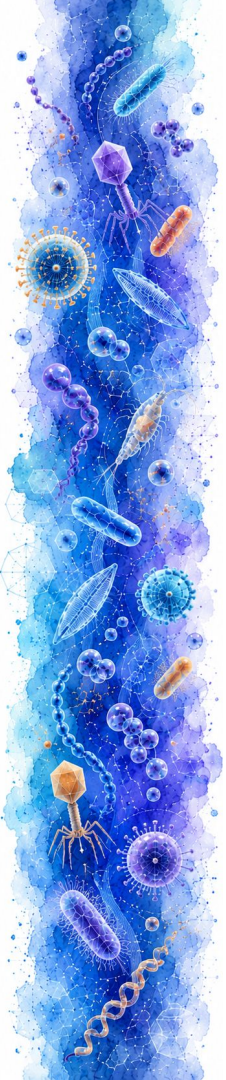


REDUCTION



Acetate + Sulphate + Hydrogen $\longrightarrow$ Carbon Dioxide + Hydrogen Sulfide + Water





This Week Read

Hug LA, Baker BJ, Anantharaman K, et al. A new view of the tree of life. *Nat Microbiol.* 2016;1(5):1-6. doi:10.1038/nmicrobiol.2016.48